# SpikeFSL: Spike-Driven Few-Shot Learning for Cross-Domain Hyperspectral Image Classification

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***Abstract*—Few-shot learning (FSL) has become an effective paradigm for cross-domain hyperspectral image (HSI) classification. However, most existing FSL methods are based on artificial neural networks (ANNs), which tend to learn domain-specific statistical representations, thereby exhibiting limited cross-domain generalization and high computational costs that restrict their applicability on resource-constrained platforms. To address these limitations, we leverage the sparse, event-driven properties of spiking neural networks (SNNs) and propose a Spike-Driven FSL (SpikeFSL) framework for cross-domain HSI classification. Specifically, we first introduce a spectral–spatial–temporal gated spiking encoding (SSTGSE) module to transform HSI pixels into SNN-compatible spiking sequences with spike-driven attentional selectivity. Then, we propose a lightweight spiking–convolutional feature extraction network (LSCFEN), which integrates spiking convolution blocks with 1-D spectral convolutions for efficient and generalizable feature learning. In addition, we develop an imitation-to-focus prototype refinement (IFPR) strategy to enhance cross-domain robustness by enabling mutual imitation between the representational strengths of SNNs and ANNs and leveraging the resulting similarity scores to focus on hard-negative samples during prototype refinement. Extensive experiments on four benchmark datasets demonstrate that SpikeFSL achieves superior accuracy and substantially lower computational cost compared with state-of-the-art FSL methods. The source code will be made publicly available at https://github.com/liang-conqueror/SpikeFSL**

***Index Terms*—Cross-domain hyperspectral image (HIS) classification, few-shot learning (FSL), prototype refinement, spiking neural networks (SNNs), spiking–convolutional feature extraction.**

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## I. Introduction

HYPERSPECTRAL images (HSIs) are capable of capturing the spatial structures and spectral information of land-cover areas, providing a powerful means to distinguish subtle differences among ground objects [1]. Consequently, HSIs have become indispensable in a wide range of practical applications, including ecological monitoring, precision agriculture, emergency response, and national security [2], [3]. As the demand for accurate scene interpretation continues to increase, HSI classification [4], [5], [6] has gradually emerged as a central topic in HSI processing. For instance, Liu et al. [7] introduced MATA to address object size variations in HSI classification via multiarea attention and later proposed SegHSI [8], an end-to-end HSI classification model for efficient inference. Zhang et al. [9] proposed TSTNet, which uses graph convolutional networks to model topological structures and address the spectral shift problem in HSI classification. However, due to complex imaging conditions and the high cost of annotation, labeled HSI samples are extremely scarce, which severely restricts the performance of traditional supervised models. Therefore, leveraging cross-domain information to alleviate the problem of insufficient labels has attracted increasing attention, and cross-domain HSI classification has thus become an important research topic. Specifically, cross-domain HSI classification seeks to transfer knowledge from a source domain with abundant labeled samples to a target domain with only limited labeled samples and significant domain shifts while learning transferable representations that generalize robustly under limited supervision.

The core idea of few-shot learning (FSL) [10] is highly consistent with this goal, as it effectively utilizes abundant labeled data from the source domain to learn generalized representations for cross-domain adaptation. Without requiring a large number of labeled target-domain samples, the model can quickly adapt to new classification tasks with only a small number of labeled samples, thereby efficiently addressing the problem of scarce annotations in cross-domain scenarios. Therefore, FSL has been widely applied in cross-domain HSI classification tasks. For example, deep cross-domain FSL (DCFSL) [11] is a representative FSL method based on convolutional neural networks (CNNs) for cross-domain HSI classification. However, existing FSL methods for cross-domain HSI classification predominantly rely on artificial neural networks (ANNs). Due to the lack of biologically inspired mechanisms and their tendency to learn

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domain-specific features, ANNs often exhibit limited cross-domain generalization capability [12], [13]. In addition, ANNs are typically based on continuous-valued computations, which makes them more sensitive to input noise and thus less robust. Meanwhile, their relatively high computational complexity further limits their deployment on resource-constrained remote sensing platforms [14], [15].

In recent years, spiking neural networks (SNNs), often regarded as the third generation of neural networks, have attracted increasing attention due to their brain-inspired computing paradigm characterized by event-driven processing, high sparsity, and low computational complexity. By mimicking the neural information processing mechanisms of the biological brain [16], [17], SNNs offer new perspectives for addressing the aforementioned challenges. In this context, several studies have introduced SNNs into single-domain HSI classification and achieved promising results. For example, Datta et al. [18] first explored the application of SNNs to single-domain HSI classification. In addition, Liu et al. [19] proposed a brain-inspired SNN based on leaky integrate-and-fire (LIF) neurons and direct encoding for single-domain HSI classification. However, existing SNN-based studies on HSI classification have mainly focused on single-domain scenarios, and efficient SNN-based FSL frameworks tailored to cross-domain HSI classification remain largely unexplored.

To fill the research gap of SNNs in cross-domain HSI classification and address the limitations of existing ANN-based FSL methods, such as high computational complexity and limited robustness under domain shifts, this article proposes a spike-driven FSL (SpikeFSL) framework. By leveraging the inherent advantages of SNNs, including event-driven processing and lightweight computation, the proposed method is designed to improve cross-domain generalization while maintaining computational efficiency, making it suitable for deployment on resource-constrained edge devices. Specifically, the main contributions of this article are as follows.

1) To the best of our knowledge, this is the first work to introduce an SNN-based framework for cross-domain HSI classification, providing a new paradigm for addressing the challenges of domain discrepancy and high computational complexity in cross-domain HSI classification.
2) We propose a spectral–spatial–temporal gated spiking encoding (SSTGSE) module to effectively transform HSI pixels into SNN-compatible spiking sequences, enabling joint spectral–spatial–temporal information modeling with spike-driven attentional selectivity.
3) We design a lightweight spiking–convolutional feature extraction network (LSCFEN) that integrates spiking convolutional blocks with 1-D spectral convolutions, enabling low computational cost while maintaining strong feature extraction capability for efficient and generalizable feature learning in cross-domain few-shot scenarios.
4) We propose an imitation-to-focus prototype refinement (IFPR) strategy, which first employs centered kernel alignment (CKA) between SNN and CNN to enable mutual imitation of their distinct representational strengths. The resulting similarity scores are then used to focus on hard-negative samples for prototype refinement, thereby improving cross-domain robustness.
5) Extensive experiments on multiple benchmark datasets demonstrate that the proposed SpikeFSL framework achieves superior classification performance with significantly lower computational cost compared to state-of-the-art FSL methods (see Fig. 1).

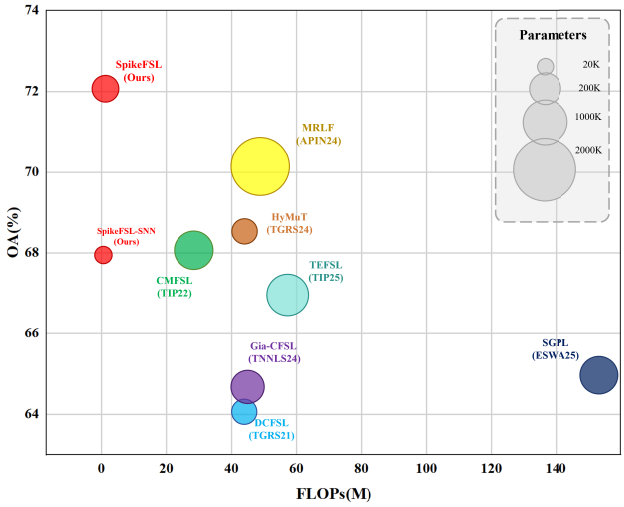



Fig. 1. Comparative analysis of OA (%), FLOPs, and model parameters among representative FSL methods on the IP dataset. The horizontal axis denotes FLOPs, and the circle size reflects the number of parameters. The proposed SpikeFSL achieves the best accuracy while maintaining the lowest computational complexity among the compared methods. SpikeFSL-SNN denotes a variant of our model that retains only the spike convolution block branch in LSCFEN.

## II. Related Work

### A. Few-Shot Learning

FSL enables models to generalize across domains by extracting transferable representations or adaptive priors from source data, requiring only minimal labeled samples from the target domain. Current FSL paradigms are primarily categorized into meta-learning [20] and metric learning [21]. Meta-learning approaches capture task-agnostic priors via episodic training, empowering models to rapidly adapt to unseen tasks. Model-agnostic meta-learning (MAML), proposed by Finn et al. [20], is a representative optimization-based method that optimizes model initialization, allowing rapid convergence on new tasks with only a few gradient updates. In contrast, metric learning focuses on constructing discriminative embedding spaces by optimizing feature manifolds and distance metrics to achieve compact intraclass aggregation. Prototypical networks, introduced by Snell et al. [22], exemplify this idea by mapping inputs into a metric space where each class is represented by the prototype of its support set, and classification is performed via distance similarity.

Collectively, meta-learning and metric learning enhance model robustness under data scarcity through distinct mechanisms and have been extensively applied to cross-domain tasks in medical imaging and remote sensing [23], [24], [25].

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These advancements provide a solid foundation for cross-domain HSI classification. In cross-domain HSI classification tasks, the FSL task is typically formulated as an episode-based $N$-way $K$-shot classification setting, where $N$ denotes the number of classes in each episode and $K$ denotes the number of labeled samples per class in the support set. In each episode, a support set and a query set are constructed, and cross-domain prototype-matching classification is conducted in a metric-learning manner. Building upon the FSL paradigm, this article explores spike-driven improvements to better address the unique challenges of spectral–spatial feature learning in cross-domain few-shot scenarios.

### B. Cross-Domain HSI Classification

HSI classification aims to exploit rich spectral–spatial information for precise pixel-level identification of land-surface objects [26], [27]. Early approaches primarily relied on traditional machine learning algorithms, such as support vector machines (SVMs) and random forests [28], which perform classification based on manually engineered spectral features. To overcome the limitations of shallow representations and poor generalization, Zhong et al. [29] proposed the spectral–spatial residual network (SSRN), establishing an end-to-end 3-D deep framework that eliminates feature engineering. However, the performance of these deep models degrades significantly when constrained by the scarcity of labeled HSI samples.

To mitigate the reliance on massive labeled data, research focus has shifted toward cross-domain HSI classification, which transfers knowledge from data-rich source domains to target domains. Liu et al. [10] pioneered Deep FSL (DFSL). It integrates residual 3D-CNNs with metric learning to construct a generalized spectral–spatial metric space via episodic training, thereby alleviating the sample scarcity bottleneck. To further tackle domain shift, Li et al. [11] proposed DCFSL, a unified framework that aligns domain distributions via conditional adversarial domain adaptation. Additionally, Xi et al. [30] developed an FSL framework with a class-covariance metric (CMFSL), which models class distributions using Mahalanobis distance to enhance accuracy while reducing computational complexity in few-shot scenarios.

Recent advancements seek further accuracy improvements by refining feature interactions. Zhang et al. [31] proposed a graph information aggregation framework (Gia-CFSL), which mitigates domain shift through feature- and distribution-level graph alignment. In addition, Zhang et al. [32] proposed BiDA, a three-branch transformer-based method to achieve bidirectional domain adaptation, further enhancing cross-domain HSI classification. Feng et al. [33] proposed a cross-domain FSL method based on decoupled knowledge distillation (DKD-FSL), which integrates metaknowledge extraction and debiasing through decoupled distillation for cross-domain HSI classification. They later introduced a transformer-based cross-domain FSL (TCFSL) method [34], which performs feature- and distribution-level adaptation for hyperspectral data under prior scarcity and class imbalance. Addressing the underexploitation of sample relationships, Liu et al. [35] introduced a multilevel relation learning framework (MRLF) to extract discriminative features. Moreover, Liu et al. [36] proposed a hybrid multitask learning (HyMuT) framework to enhance cross-domain HSI classification performance through multiview feature extraction. More recently, Feng et al. [37] proposed a cross-domain FSL method based on fractional domain information (FrCFSL), which leverages the fractional Fourier transform to extract fractional features and effectively alleviates semantic cross-coupling in HIS analysis. Li et al. [38] proposed semantic-guided prototype learning (SGPL), which generates more discriminative prototypes by fusing semantic label information, while Xi et al. [39] presented a Transductive Enhanced FSL (TEFSL) framework to enable test-time adaptation. Despite these strides, existing methods predominantly rely on traditional ANNs. However, ANNs lack the biological plausibility of brain-inspired mechanisms, which may limit their robustness in complex HSI scenarios, while their continuous-valued computations often result in relatively low computational efficiency.

### C. Spiking Neural Network

SNNs employ discrete spikes and an event-driven computation paradigm to emulate biological information processing in the brain, which fundamentally differs from the continuous-valued computations of ANNs. Their neural dynamics enable greater stability under noise perturbations and distribution shifts [40], [41]. These advantages have motivated SNN-based applications across multiple research domains. In FSL, Gholamzadeh Khoee et al. [42] proposed a biologically plausible SNN meta-learning model to alleviate catastrophic forgetting. For cross-domain problems, Zhan et al. [41] developed a CKA-based deep SNN transfer learning method, verifying the transferability of multilayer features in deep SNNs.

SNNs have also been applied to single-domain HSI classification. Liu et al. [19] proposed a brain-inspired SNN framework based on LIF neurons and direct encoding, achieving efficient spectral–spatial feature extraction using spiking depthwise separable convolutions. Li et al. [43] further introduced an SNN framework with adaptive refocusing convolutional layers and a spike self-attention mechanism, improving feature extraction under short time-step conditions. Despite these advances, the application of SNNs to cross-domain HSI classification remains largely unexplored. Due to pronounced domain shifts, class imbalance, and spatial–spectral distribution discrepancies, SNNs designed for single-domain tasks cannot be directly applied to cross-domain scenarios. Jiao et al. [44] systematically reviewed brain-inspired learning algorithms from neural microstructures to cognitive behaviors, highlighting sparsity, selectivity, and plasticity as key inspirations for next-generation artificial intelligence. Motivated by these insights, this study explores a hybrid SNN–ANN architecture, constructing a SpikeFSL framework for cross-domain HSI classification.

## III. Proposed Method

### A. Overall Architecture

This article proposes a SpikeFSL framework for cross-domain HSI classification, termed SpikeFSL. As illustrated

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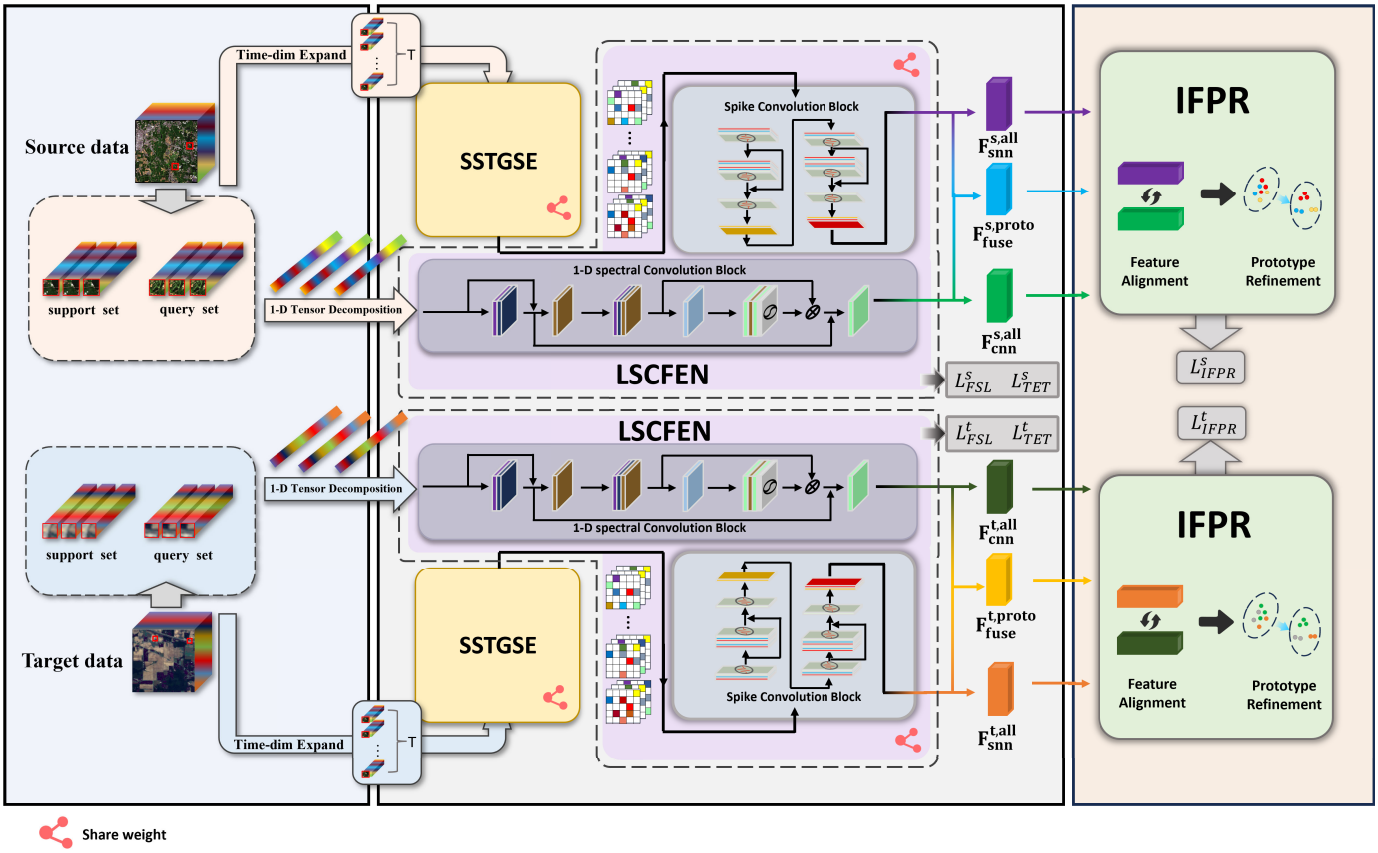



Fig. 2. Framework of SpikeFSL.

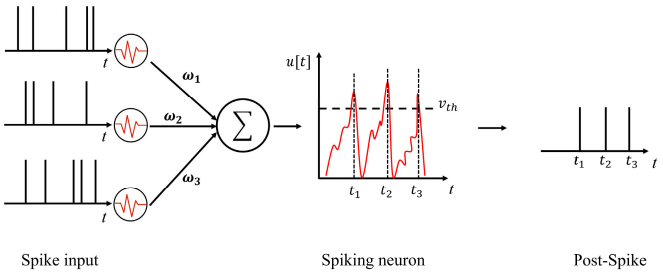



Fig. 3. Illustration of the LIF neuron dynamics used in the SNN.

in Fig. 2, the framework consists of three main components corresponding to the proposed innovations. First, the SSTGSE module converts HSI patches into informative spike sequences for SNN-compatible processing. Second, an LSCFEN extracts features via parallel SNN and CNN branches and fuses the resulting representations. Finally, an IFPR strategy is applied to enhance cross-domain robustness. In Fig. 2, $\mathbf{F}^{\mathbf{s},\text{all}}_{\text{snn}}$, $\mathbf{F}^{\mathbf{s},\text{all}}_{\text{cnn}}$, and $\mathbf{F}^{\mathbf{s},\text{proto}}_{\text{fuse}}$ denote the SNN features, CNN features, and fused prototype features from the source domain, respectively, while $\mathbf{F}^{\mathbf{t},\text{all}}_{\text{snn}}$, $\mathbf{F}^{\mathbf{t},\text{all}}_{\text{cnn}}$, and $\mathbf{F}^{\mathbf{t},\text{proto}}_{\text{fuse}}$ denote the corresponding features from the target domain.

### B. LIF Neuron Model

SNNs use spike trains as carriers to emulate biological information transmission mechanisms and neuronal dynamics. Among various neuron models, the LIF model [45] is widely adopted due to its favorable tradeoff between biological plausibility and computational efficiency. As illustrated in Fig. 3, an LIF neuron integrates incoming spikes over time to accumulate the membrane potential and emits a spike once the potential exceeds a predefined threshold. This process consists of three stages: membrane potential integration, spike generation, and membrane potential reset. The discrete-time formulation of the LIF neuron is as follows:

$$u[t] = v[t-1] + \frac{1}{\xi}\left(x[t] - (v[t-1] - v_{\text{reset}})\right) \tag{1}$$

$$O[t] = H(u[t] - v_{\text{th}}) = \begin{cases} 1, & u[t] \geq v_{\text{th}} \\ 0, & u[t] < v_{\text{th}} \end{cases} \tag{2}$$

$$v[t] = u[t](1 - O[t]) + v_{\text{reset}} O[t] \tag{3}$$

where $x[t]$ denotes the synaptic input at time step $t$; $u[t]$ represents the membrane potential before spike emission; $v[t]$ denotes the membrane potential after spike emission; $\xi$ is the membrane time constant, $v_{\text{reset}}$ and $v_{\text{th}}$ denote the reset potential and firing threshold, respectively; and $H(\cdot)$ is the Heaviside step function, which outputs 1 when $u[t] \geq v_{\text{th}}$ and 0 otherwise.

Since the step function in (2) is nondifferentiable, LIF neurons cannot be trained directly via standard backpropagation. The surrogate gradient method [46] is therefore adopted. In the forward pass, spikes are generated by (2), and in the backward

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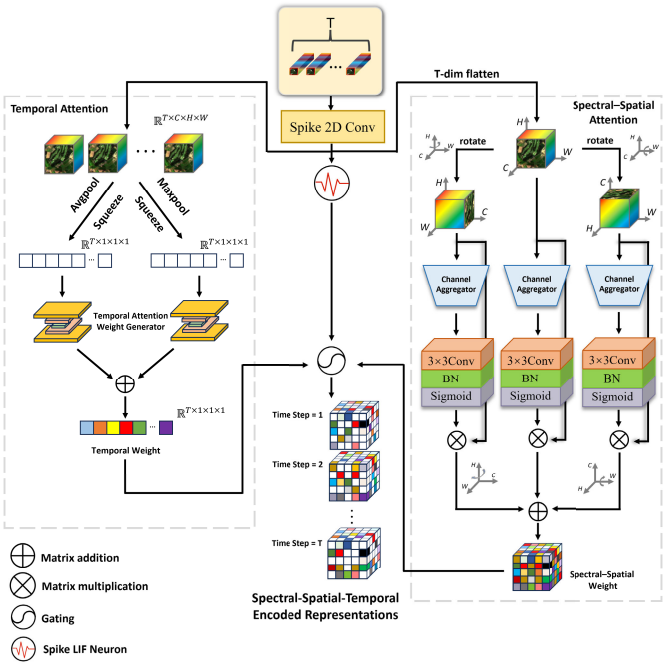



Fig. 4. Architecture of the SSTGSE module, consisting of a TA block and an SSA block for joint spectral–spatial–temporal modeling.

pass, the derivative of the spike function is approximated using that of the sigmoid function

$$\frac{\partial O[t]}{\partial u[t]} \approx \alpha \cdot \text{sigmoid}(\alpha u[t]) \cdot \left(1 - \text{sigmoid}(\alpha u[t])\right) \tag{4}$$

where $\text{sigmoid}(\alpha u[t]) = 1/(1 + e^{-\alpha u[t]})$ and $\alpha$ is a hyperparameter controlling the smoothness of the surrogate gradient.

### C. Spectral–Spatial–Temporal Gated Spiking Encoding

To accommodate the temporal computing paradigm of SNNs based on LIF neurons, static HSI data need to be expanded along the temporal dimension. Inspired by [47], a strategy is adopted that replicates the original data $T$ times, expanding the static input to $T$ time steps to construct the initial temporal input. This method requires no additional parameters, incurs low computational cost, and preserves the original features, providing a clean input representation for subsequent processing.

Next, the $T$-dimensional temporal data are encoded to suppress the redundancy introduced by the replication method. Most existing SNNs adopt a direct-encoding strategy [48], which maps input data into binary spike sequences through simple convolutional projections and LIF neuron firing. While lightweight and efficient, such hard encoding treats all input information uniformly and often leads to the loss of fine-grained spectral–spatial details, which are critical for HSI analysis. To address these limitations, we propose an SSTGSE module. As illustrated in Fig. 4, SSTGSE jointly models temporal dynamics and spectral–spatial structures by incorporating a temporal attention (TA) block and a spectral–spatial attention (SSA) block. Through a multipath selective gating mechanism, SSTGSE refines the constructed temporal information and suppresses redundant spectral–spatial information while selectively enhancing salient responses, thereby generating informative spiking representations.

For the TA module, the objective is to assign differentiated weights to features across $T$ time steps, thereby focusing on time-step features that contain key discriminative information. The input HSI patch is first fed into a spiking 2-D convolution block to extract preliminary features, resulting in a spectral–spatial–temporal representation $\mathbf{X} \in \mathbb{R}^{T\times C\times H\times W}$, where $T$ denotes the temporal length obtained by replicating the original data, $H$ and $W$ are the spatial dimensions, and $C$ is the number of spectral channels, empirically set to 100. After this spiking convolution operation, the features across different time steps become differentiated, enabling the extraction of informative temporal representations.

To capture global temporal statistics, 3-D adaptive average pooling and max pooling are applied to aggregate spatial and channel information

$$\mathbf{X}_{\text{avg}} = \text{AvgPool}_{3D}(\mathbf{X}), \quad \mathbf{X}_{\text{max}} = \text{MaxPool}_{3D}(\mathbf{X}) \tag{5}$$

where $\mathbf{X}_{\text{avg}}, \mathbf{X}_{\text{max}} \in \mathbb{R}^{T\times 1\times 1\times 1}$.

The pooled descriptors are then fed into a shared TA weight generator composed of two pointwise convolutions with a ReLU activation, which models temporal dependencies via dimensionality reduction and expansion

$$\mathbf{A}_i = \mathbf{W}_2(\delta(\mathbf{W}_1(\mathbf{X}_i))), \quad i \in \{\text{avg}, \text{max}\} \tag{6}$$

where $\mathbf{A}_{\text{avg}}, \mathbf{A}_{\text{max}} \in \mathbb{R}^{T\times 1\times 1\times 1}$, $\delta(\cdot)$ denotes the ReLU function, and $\mathbf{W}_1$ and $\mathbf{W}_2$ represent pointwise convolution operators.

Finally, the outputs from the average- and max-pooling branches are combined to obtain the TA weights

$$\mathbf{A}_{\text{TA}} = \mathbf{A}_{\text{avg}} + \mathbf{A}_{\text{max}} \tag{7}$$

where $\mathbf{A}_{\text{TA}} \in \mathbb{R}^{T\times 1\times 1\times 1}$. The resulting attention weights enable differentiated modulation of spatial features across time steps, allowing each time step to emphasize distinct discriminative regions.

For the SSA module, the objective is to capture joint dependencies between the spectral and spatial dimensions. Specifically, SSA is composed of three parallel branches that model complementary interactions along the height–channel ($H$–$C$), channel–width ($C$–$W$), and height–width ($H$–$W$) dimensions. The input to SSA is the spectral–spatial–temporal feature tensor $\mathbf{X}$ obtained in the previous stage. A temporal flattening operation is first applied to aggregate temporal information, resulting in a spectral–spatial representation $\mathbf{X}' \in \mathbb{R}^{C\times H\times W}$.

In the height–channel ($H$–$C$) branch, $\mathbf{X}'$ is permuted to align the height and channel dimensions

$$\mathbf{X}'_{hc} = \text{permute}\left(\mathbf{X}', (W, H, C)\right). \tag{8}$$

The permuted features are then processed by a channel aggregator (CA) module, which computes both the channel-wise maximum and mean at each spatial location and concatenates them

$$\text{CA}\left(\mathbf{X}'_{hc}\right) = \text{Concat}\left(\max_c \mathbf{X}'_{hc}, \frac{1}{C}\sum_c \mathbf{X}'_{hc}\right) \tag{9}$$

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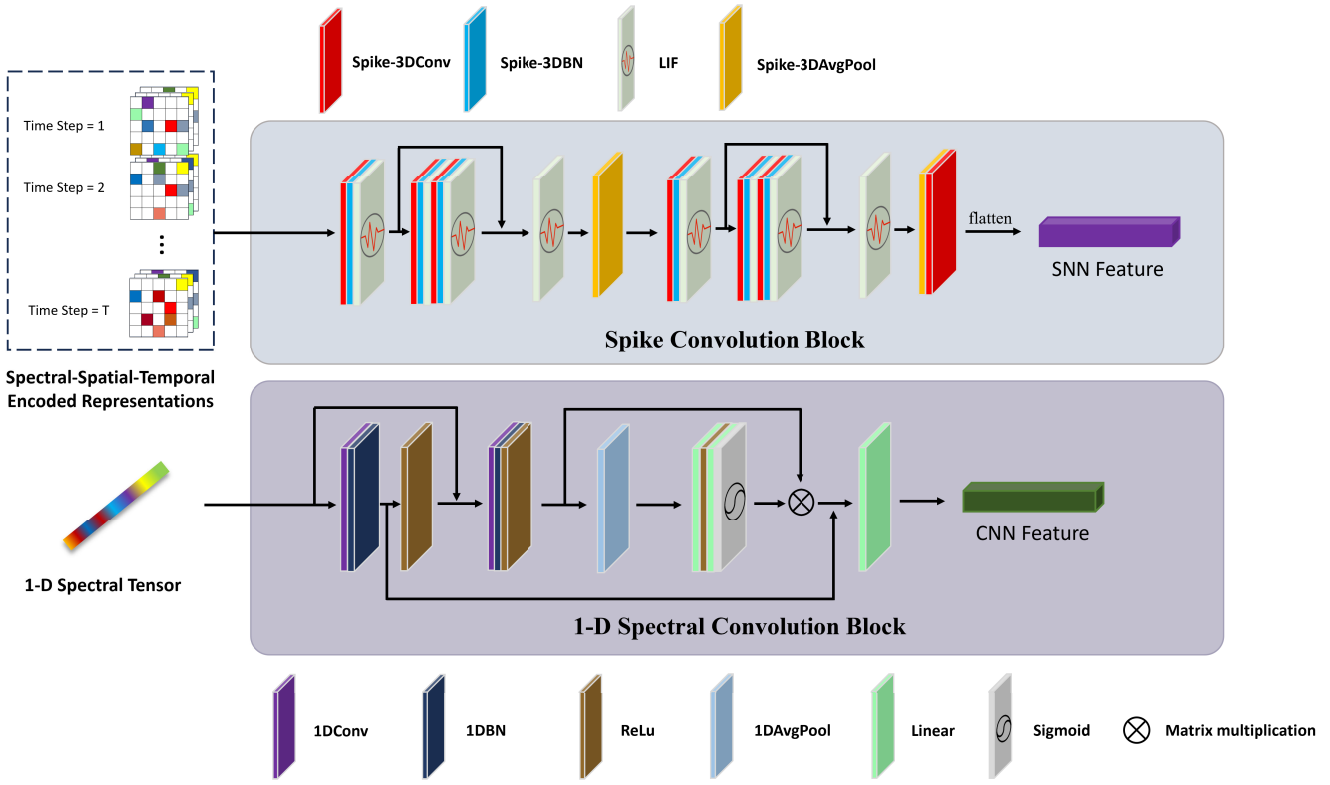



Fig. 5. Architecture of the LSCFEN, which integrates a spiking convolution block and a 1-D spectral convolution block to jointly extract features under the SNN and CNN paradigms.

where $\mathrm{CA}(\mathbf{X}'_{hc}) \in \mathbb{R}^{2\times H\times C}$. Note that the aggregated dimension corresponds to the permuted channel axis, which may represent different physical dimensions in different branches.

The aggregated features are further transformed to generate attention weights

$$\omega_{hc} = \sigma\left(\mathrm{BN}\left(f_{\mathrm{conv3\times3}}\left(\mathrm{CA}\left(\mathbf{X}'_{hc}\right)\right)\right)\right) \tag{10}$$

where $f_{\mathrm{conv3\times3}}(\cdot)$ denotes a standard $3 \times 3$ convolution, $\mathrm{BN}(\cdot)$ is batch normalization, $\sigma(\cdot)$ is the sigmoid activation function, and $\omega_{hc} \in \mathbb{R}^{1\times H\times C}$.

The attention weights are applied to the permuted features via element-wise multiplication and then rearranged back

$$\mathbf{A}_{hc} = \mathrm{permute}\left(\omega_{hc} \odot \mathbf{X}'_{hc}, (C, H, W)\right) \tag{11}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{A}_{hc} \in \mathbb{R}^{C\times H\times W}$.

The same procedure is applied to the channel–width ($C$–$W$) and height–width ($H$–$W$) branches, yielding $\mathbf{A}_{cw}$ and $\mathbf{A}_{hw}$, respectively. The outputs of the three branches are averaged to obtain the final SSA feature map

$$\mathbf{A}_{\mathrm{SSA}} = \frac{1}{3}\left(\mathbf{A}_{hc} + \mathbf{A}_{cw} + \mathbf{A}_{hw}\right). \tag{12}$$

The temporal dimension is then reintroduced by broadcasting $\mathbf{A}_{\mathrm{SSA}}$ along the time axis, resulting in $\mathbf{A}_{\mathrm{SSA}} \in \mathbb{R}^{T\times C\times H\times W}$.

Finally, given the spike tensor $\mathbf{Spike} \in \{0,1\}^{T\times C\times H\times W}$ obtained from the spectral–spatial–temporal representation $\mathbf{X}$ via LIF neurons, the gated spiking encoding is formulated as

$$\mathbf{X}_{\mathrm{out}} = \sigma\left(\mathbf{A}_{\mathrm{TA}} \odot \mathbf{A}_{\mathrm{SSA}}\right) \odot \mathbf{Spike}. \tag{13}$$

The resulting $\mathbf{X}_{\mathrm{out}} \in \mathbb{R}^{T\times C\times H\times W}$ is a selectively modulated spiking representation, where zero values indicate complete suppression and nonzero continuous values correspond to different levels of attentional emphasis, thereby enabling spike-driven attentional selectivity in the encoding process. It is worth noting that the output of this module consists of zero and nonzero continuous values. Therefore, the computational advantage of the model does not stem from this module, but mainly from the subsequent LSCFEN.

### *D. Lightweight Spiking–Convolutional Feature Extraction Network*

The proposed LSCFEN is motivated by the complementary characteristics of CNNs and SNNs. CNNs are effective in dense representation learning, but they often involve high computational cost and may be sensitive to redundant or noisy inputs. In contrast, SNNs employ sparse, event-driven processing, which can help suppress redundant responses and reduce computational cost, but they are generally less effective in capturing dense and fine-grained features. To leverage the strengths of both paradigms while maintaining model compactness, we design a dual-path architecture so that the fused features benefit from the computational efficiency of SNNs and the strong representation capability of CNNs. As illustrated in Fig. 5, LSCFEN consists of two parallel branches: a spike convolution block operating under the SNN paradigm and a 1-D spectral convolution block operating under the CNN paradigm, followed by feature fusion.

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*1) Spike Convolution Block:* The spike convolution block operates on the spiking representation $\mathbf{X}_{\text{out}}$ generated by the SSTGSE module. The input is first passed through a mapping layer, consistent with the HSI mapping strategy in [49], to unify the channel dimensions of the source and target domains. The core feature extractor consists of two residual spiking convolution subblocks, interleaved with spiking convolutional layers, LIF neurons, and 3-D spiking average-pooling layers. Each residual subblock comprises a $3 \times 3 \times 3$ spiking convolution, a spiking batch normalization layer, and an LIF neuron, enabling local spectral–spatial–temporal feature extraction while maintaining spike sparsity. Residual connections alleviate gradient degradation and stabilize deep spiking feature learning. Each subblock is followed by a 3-D spiking average-pooling layer to further aggregate spectral–spatial–temporal information.

The spike convolution block produces a temporal spiking feature tensor $\mathbf{F}_{\text{snn,T}}$ with $T$ time steps and a feature dimension of $N^s$. A temporally aggregated representation is then obtained by averaging over time

$$\mathbf{F}_{\text{snn}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{F}_{\text{snn,T}}[t,:,:,:] \tag{14}$$

where $\mathbf{F}_{\text{snn}} \in \mathbb{R}^{N^s}$ is used for cross-paradigm feature fusion, while $\mathbf{F}_{\text{snn,T}}$ is retained for computing the temporal efficient training (TET) loss.

*2) 1-D Spectral Convolution Block:* To complement spiking-based spectral–spatial–temporal features, a 1-D spectral convolution block is introduced to capture fine-grained spectral structures. The input HSI patch is first decomposed into a 1-D spectral tensor, which extracts compact and informative spectral representations. The resulting tensor is passed through a mapping layer consisting of a 1-D convolution and batch normalization to stabilize training. Feature extraction is then performed using two small residual subblocks and one large residual subblock. The small residual subblocks capture local spectral correlations using $1 \times 3$ convolutions, while the large residual subblock integrates long-range interband dependencies through hierarchical spectral aggregation. The final spectral feature is projected into an $N^c$-dimensional vector $\mathbf{F}_{\text{cnn}} \in \mathbb{R}^{N^c}$.

*3) Cross-Paradigm Feature Fusion:* To ensure compatibility between the two branches, both $\mathbf{F}_{\text{snn}}$ and $\mathbf{F}_{\text{cnn}}$ are aligned to the same dimensionality and fused by averaging

$$\mathbf{F}_{\text{fuse}} = \frac{\mathbf{F}_{\text{snn}} + \mathbf{F}_{\text{cnn}}}{2}. \tag{15}$$

In our implementation, both $N^s$ and $N^c$ are set to 160. This fusion strategy integrates the sparse response characteristics of SNN features with the fine-grained spectral modeling capability of CNN features, yielding a lightweight and discriminative representation that is well-suited for cross-domain HSI classification.

### *E. Imitation-to-Focus Prototype Refinement*

Although the proposed LSCFEN effectively integrates spiking and convolutional representations, the SNN and CNN paradigms inherently rely on distinct encoding mechanisms. Specifically, SNN features are characterized by sparse, event-driven activations, whereas CNN features exhibit dense, continuous representations. This discrepancy leads to an intrinsic misalignment between their feature manifolds, particularly in their feature distributions, as illustrated in Fig. 6(b). Without additional structural constraints, such heterogeneity may hinder the effective exploitation of their complementary strengths under domain shift.

Nevertheless, from a semantic perspective, both paradigms encode the same underlying sample identities, a property evidenced by the strong diagonal consistency in the sample-level similarity heatmap shown in Fig. 6(a). This observation suggests that, although the two branches may differ substantially in feature distribution and geometry, they still preserve related semantic relationships among samples. Therefore, the key objective is not to force the two branches to share an identical feature distribution, but to maintain consistent semantic structure across heterogeneous representations. Moreover, in few-shot settings, class prototypes often suffer from blurred interclass boundaries due to limited samples, further hindering generalization across domains.

To address these challenges, we propose an IFPR strategy. The proposed strategy first aligns the semantic geometry across different paradigms via CKA between SNN and CNN, enabling mutual imitation of their distinct representational strengths. It then uses the resulting similarity scores to focus on hard-negative samples for prototype refinement, thereby enhancing cross-domain robustness.

*1) CKA Between SNN and CNN:* Consider a training batch where the support set, query set, and cross-domain samples are concatenated along the sample dimension following the protocol in [11]. Let the concatenated SNN- and CNN-path features be:

$$\mathbf{F}_{\text{snn}}^{\text{s,all}} = \text{cat}\left(\mathbf{F}_{\text{snn}}^{s,p}, \mathbf{F}_{\text{snn}}^{s,q}, \mathbf{F}_{\text{snn}}^{\text{target}}\right) \in \mathbb{R}^{B\times N^s} \tag{16}$$

$$\mathbf{F}_{\text{cnn}}^{\text{s,all}} = \text{cat}\left(\mathbf{F}_{\text{cnn}}^{s,p}, \mathbf{F}_{\text{cnn}}^{s,q}, \mathbf{F}_{\text{cnn}}^{\text{target}}\right) \in \mathbb{R}^{B\times N^c} \tag{17}$$

where $B$ denotes the total number of concatenated samples and $N^s$ and $N^c$ denote the feature dimensions of the SNN and CNN branches, respectively.

To stabilize the alignment process and suppress noise, both feature streams are projected into a shared low-dimensional subspace using learnable projection heads

$$\mathbf{Z}_{\text{snn}} = P_s\left(\mathbf{F}_{\text{snn}}^{\text{s,all}}\right), \quad \mathbf{Z}_{\text{cnn}} = P_c\left(\mathbf{F}_{\text{cnn}}^{\text{s,all}}\right) \tag{18}$$

where $P_s$ and $P_c$ denote the projection heads for the SNN and CNN paths, respectively, and $\mathbf{Z}_{\text{snn}}, \mathbf{Z}_{\text{cnn}} \in \mathbb{R}^{B\times d}$ with $d < \min(N^s, N^c)$.

Using a linear kernel, the Gram matrix of a feature matrix $\mathbf{Z}$ is defined as

$$\mathbf{G} = \mathbf{Z}\mathbf{Z}^\top, \quad \mathbf{G}_{ij} = \mathbf{Z}_i^\top \mathbf{Z}_j \tag{19}$$

where $\mathbf{Z}_i$ denotes the $i$th sample feature vector.

To remove the influence of feature means, the Gram matrix is centered before computing similarity. Specifically, given a Gram matrix $\mathbf{G} \in \mathbb{R}^{B\times B}$, its centered version is obtained as

$$\tilde{\mathbf{G}} = \mathbf{H}\mathbf{G}\mathbf{H}, \quad \mathbf{H} = \mathbf{I} - \frac{1}{B}\mathbf{1}\mathbf{1}^\top \tag{20}$$

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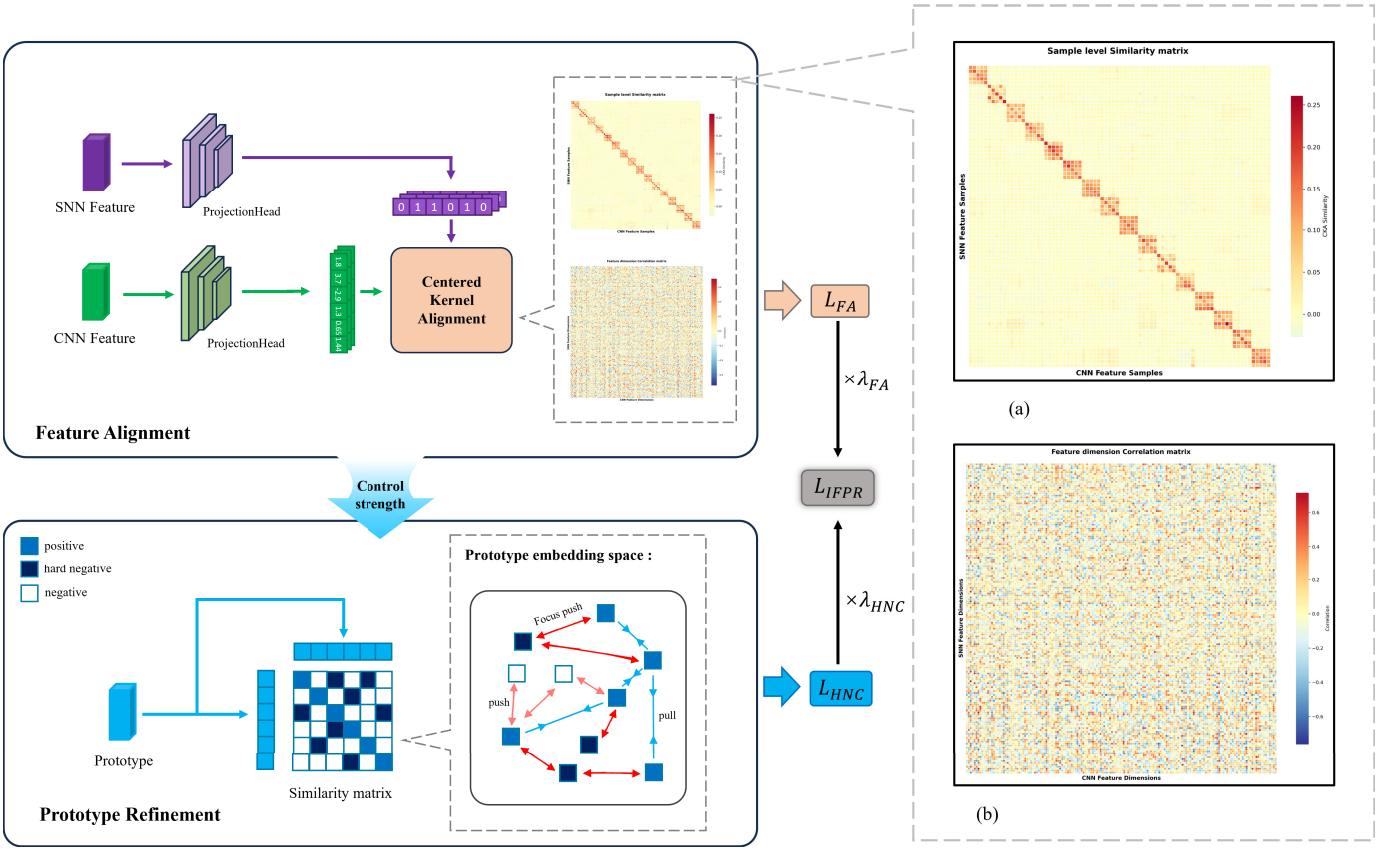



Fig. 6. Illustration of the proposed IFPR. (a) Sample-level CKA is used to align the representations of SNN and CNN, ensuring consistent semantic alignment across different network paradigms. (b) Correlation analysis across feature dimensions reveals geometric mismatch within the feature space.

where $\mathbf{I}$ is the identity matrix and $\mathbf{1}$ is an all-ones vector.

Based on the centered Gram matrices, the statistical dependence between two representations is quantified using the Hilbert–Schmidt independence criterion (HSIC). Given two feature representations $\mathbf{Z}_a$ and $\mathbf{Z}_b$, HSIC with a linear kernel is defined as

$$\mathrm{HSIC}(\mathbf{Z}_a, \mathbf{Z}_b) = \frac{1}{(B-1)^2}\left\langle \widetilde{\mathbf{G}}_a, \widetilde{\mathbf{G}}_b \right\rangle_F \tag{21}$$

where $\tilde{\mathbf{G}}_a$ and $\tilde{\mathbf{G}}_b$ denote the centered Gram matrices constructed from $\mathbf{Z}_a$ and $\mathbf{Z}_b$, respectively, and $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product.

To obtain a scale-invariant similarity measure, the CKA coefficient is computed by normalizing HSIC

$$L_{\mathrm{FA}} = 1 - \frac{\mathrm{HSIC}(\mathbf{Z}_{\mathrm{snn}}, \mathbf{Z}_{\mathrm{cnn}})}{\sqrt{\mathrm{HSIC}(\mathbf{Z}_{\mathrm{snn}}, \mathbf{Z}_{\mathrm{snn}})\ \mathrm{HSIC}(\mathbf{Z}_{\mathrm{cnn}}, \mathbf{Z}_{\mathrm{cnn}})}}. \tag{22}$$

Unlike distribution-matching objectives such as maximum mean discrepancy (MMD) or local MMD (LMMD), which mainly reduce discrepancy by aligning global statistics across feature spaces, the proposed CKA-based objective emphasizes the consistency of intersample relational geometry. This property is particularly important in our method, because the SNN and CNN branches observe the same samples but represent them in different forms. Therefore, minimizing the feature alignment loss $L_{\mathrm{FA}}$ encourages the SNN and CNN representations to imitate each other at the level of global semantic geometry, while preserving their intrinsic encoding properties.

*a) Hard-negative guided prototype refinement:* To further enhance prototype discriminability, we introduce a hard-negative guided prototype refinement mechanism at the prototype level. Let the fused prototype representations be defined as

$$\mathbf{F}^p_{\mathrm{fuse}} = \left\{\mathbf{F}^p_{\mathrm{fuse},i}\right\}_{i=1}^{M} \in \mathbb{R}^{M \times N^p} \tag{23}$$

where $M$ is the number of class prototypes in the current episode and $N^p$ is the prototype feature dimension. $\mathbf{F}^p_{\mathrm{fuse},i}$ denotes the prototype of the $i$th class, obtained by averaging the features of support-set samples belonging to that class. Row-wise $\ell_2$ normalization is then applied to obtain $\tilde{\mathbf{F}}_i = \mathbf{F}^p_{\mathrm{fuse},i} / \|\mathbf{F}^p_{\mathrm{fuse},i}\|_2$.

Based on the normalized prototypes, pairwise similarities are computed as

$$S_{ij} = \exp\left(\frac{\tilde{\mathbf{F}}_i^{\top} \tilde{\mathbf{F}}_j}{\tau}\right) \tag{24}$$

where $\tau > 0$ is a temperature parameter controlling the sharpness of the similarity distribution. For the $i$th anchor prototype, the positive similarity is defined as $S_{ii}$, and the negative set is $\{S_{ij} \mid j \neq i\}$. Since each prototype is $\ell_2$-normalized, $S_{ii} = \exp(1/\tau)$ is a constant for a fixed $\tau$. Here, $S_{ii}$ serves as a fixed positive anchor in the normalized contrastive loss. Both positive and negative similarities are calculated through the same exponential mapping and measured

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on a unified temperature-controlled scale. The adaptive hard-negative emphasis is introduced by the reweighted negative term defined next.

To emphasize hard negatives, i.e., negatives with high similarity to the anchor prototype, a power-based reweighting scheme is adopted

$$w_{ij} = S_{ij}^{\beta}, \quad j \neq i,\ \beta \geq 0 \tag{25}$$

where $\beta$ controls the strength of hard-negative emphasis: a larger $\beta$ assigns higher weights to more similar (harder) negatives. Moreover, $\beta$ is adaptively adjusted by the CKA coefficient computed from $\mathbf{Z}_{\text{snn}}$ and $\mathbf{Z}_{\text{cnn}}$, thereby adjusting the model's focus on hard-negative samples. Specifically, $\beta$ is defined as follows:

$$\beta = \beta_0 \cdot \frac{\text{HSIC}\left(\mathbf{Z}_{\text{snn}}, \mathbf{Z}_{\text{cnn}}\right)}{\sqrt{\text{HSIC}\left(\mathbf{Z}_{\text{snn}}, \mathbf{Z}_{\text{snn}}\right)\ \text{HSIC}\left(\mathbf{Z}_{\text{cnn}}, \mathbf{Z}_{\text{cnn}}\right)}} \tag{26}$$

where $\beta_0$ is a base hyperparameter. This formulation dynamically adjusts $\beta$ according to the CKA coefficient. Strong alignment between the feature manifolds of SNN and CNN strengthens the focus on hard negatives for prototype refinement, helping the model learn finer interclass prototype differences. In contrast, feature misalignment automatically weakens this focusing strength to prevent negative optimization caused by overly aggressive prototype refinement on poorly aligned features.

Then, the reweighted negative term for the $i$th anchor prototype is computed as

$$\text{neg}_i = \frac{\sum_{j \neq i} w_{ij} S_{ij}}{\frac{1}{M-1} \sum_{j \neq i} w_{ij}} \tag{27}$$

which preserves the overall negative magnitude while increasing the contribution from hard negatives. The per-prototype hard-negative contrastive loss is then defined as

$$\ell_i = -\log \frac{S_{ii}}{S_{ii} + \text{neg}_i} \tag{28}$$

and the overall hard-negative contrastive loss is

$$L_{\text{HNC}} = \frac{1}{M} \sum_{i=1}^{M} \ell_i. \tag{29}$$

*b) Overall optimization:* The complete IFPR strategy combines the feature alignment and hard-negative contrastive losses

$$L_{\text{IFPR}} = \lambda_{\text{FA}} L_{\text{FA}} + \lambda_{\text{HNC}} L_{\text{HNC}} \tag{30}$$

where $\lambda_{\text{FA}}$ and $\lambda_{\text{HNC}}$ balance the two loss terms.

The FSL classification loss $L_{\text{FSL}}$ is another component of the overall optimization objective [11]. Taking the source-domain training stage as an example, the metric-learning-based prototypical network paradigm is adopted, where class prototypes are constructed from the support set and the negative log probability of query samples is minimized. Concretely, in each training episode, $G$ classes are randomly selected from the source domain, with $N_1$ labeled samples per class forming the support set $\mathcal{S} = \{(x_i, y_i)\}$, and another $N_2$ samples per class drawn from the remaining data to form the query set $\mathcal{Q} = \{(x_j, y_j)\}$. Let $f_\theta(\cdot)$ denote the feature extraction network.

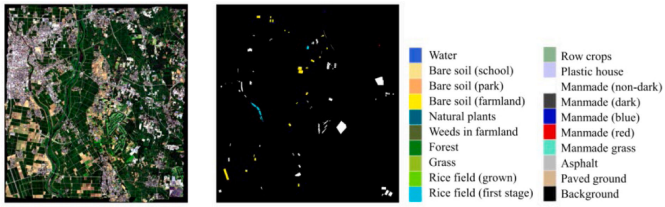



Fig. 7. Chikusei dataset: a pseudo-color image together with its corresponding ground-truth map.

The probability that a query sample $x_j$ belongs to class $k$ is defined as

$$p\left(y_j = k \mid x_j\right) = \frac{\exp\left(-\left\| f_\theta\left(x_j\right) - \mathbf{F'}^{p}_{fuse,k} \right\|^2\right)}{\sum_{k'=1}^{G} \exp\left(-\left\| f_\theta\left(x_j\right) - \mathbf{F'}^{p}_{fuse,k'} \right\|^2\right)} \tag{31}$$

where $\mathbf{F'}^{p}_{\text{fuse},k}$ denotes the prototype of the $k$th class after prototype refinement.

The FSL loss is then formulated as the expectation of the negative log likelihood over all query samples

$$L_{\text{FSL}} = \mathbb{E}_{\mathcal{S},\mathcal{Q}}\left[-\sum_{(x_j,y_j)\in\mathcal{Q}} \log p\left(y_j = k \mid x_j\right)\right]. \tag{32}$$

This loss encourages the embedding space to achieve intra-class compactness and interclass separability through metric learning. The computation of the FSL loss for the target domain follows the same procedure, except that the support and query sets are drawn from the target domain.

The TET loss $L_{\text{TET}}$ is an optimization objective specifically tailored for SNNs [50]. This loss function imposes independent classification constraints on the output at each time step to alleviate the potential mismatch between surrogate gradients and the true loss, thereby improving the temporal information utilization efficiency of the model. Formally, let $\mathbf{F}_{\text{snn,T}}[t]$ denote the network output at time step $t$, and let $y$ be the corresponding sample label. $L_{\text{TET}}$ is defined as the average of the per-time-step cross-entropy losses

$$L_{\text{TET}} = \frac{1}{T} \sum_{t=1}^{T} \mathcal{L}_{\text{CE}}\left(\mathbf{F}_{\text{snn,T}}\left[t\right], y\right) \tag{33}$$

where $T$ is the total simulation length of the SNNs and $\mathcal{L}_{\text{CE}}(\cdot)$ denotes the cross-entropy loss. In this work, $L_{\text{TET}}$ is incorporated as an auxiliary loss term into the overall optimization objective to enhance the training stability and generalizability of the spiking feature extraction module.

Finally, the total loss of the SpikeFSL framework is given by

$$L = L_{\text{IFPR}} + L_{\text{FSL}} + L_{\text{TET}}. \tag{34}$$

## IV. Experiment

### A. Dataset Description

To comprehensively evaluate the proposed SpikeFSL framework for cross-domain HSI classification, we select one publicly available dataset as the source domain and four

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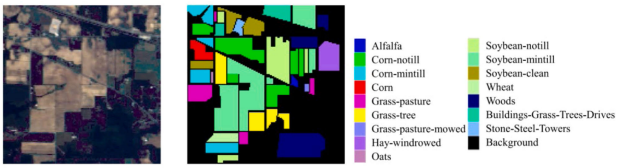



Fig. 8. IP dataset: a pseudo-color image together with its corresponding ground-truth map.

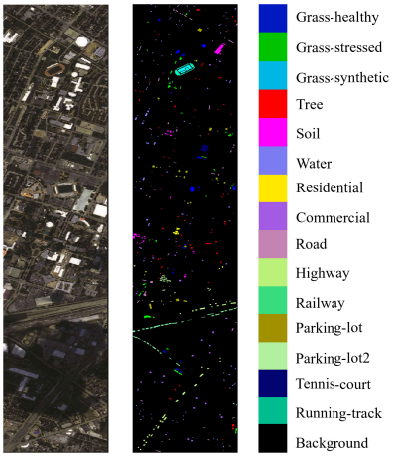



Fig. 9. HT dataset: a pseudo-color image together with its corresponding ground-truth map.

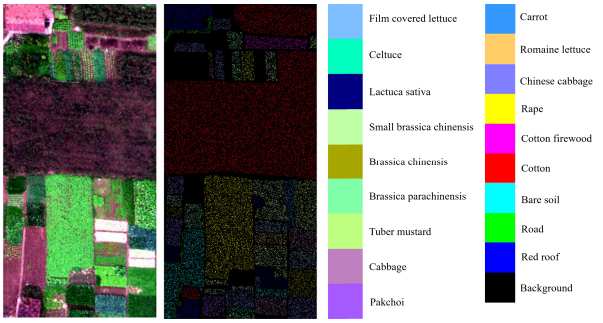



Fig. 10. HH dataset: a pseudo-color image together with its corresponding ground-truth map.

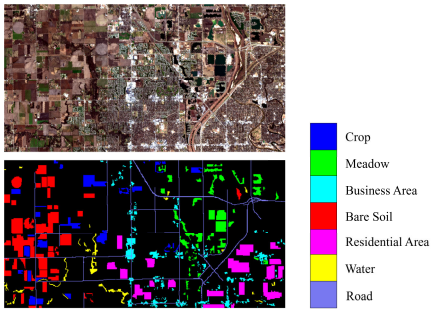



Fig. 11. Kansas dataset: a pseudo-color image together with its corresponding ground-truth map.

representative datasets as target domains. These datasets cover diverse imaging sensors, spatial resolutions, and land-cover scenarios. The source-domain dataset provides sufficient labeled samples for base-class training, while each target-domain dataset contains only a very limited number of labeled samples, thereby simulating a realistic cross-domain few-shot transfer setting. The detailed characteristics of all datasets are summarized in Table I. In addition, to provide a quantitative analysis of source–target domain discrepancy, we report the Prototypical Optimal Transport distance [51] between Chikusei and each target-domain dataset, where a larger value indicates a larger overall domain discrepancy. Corresponding pseudo-color images and ground-truth maps are presented in Figs. 7–11.

TABLE I
ALL DETAILS OF THE EXPERIMENTAL DATASETS

| Dataset | Spatial Size | Spectral Range | Band Number | Class Number | Labeled Samples | Domain Distance |
|---|---|---|---|---|---|---|
| Chikusei | 2517 × 2335 | 363–1018 nm | 128 | 19 | 77592 | - |
| IP | 145 × 145 | 400–2500 nm | 200 | 16 | 10249 | 1916 |
| HT | 349 × 1905 | 364–1046 nm | 144 | 15 | 15029 | 1371 |
| HH | 940 × 475 | 400–1000 nm | 270 | 18 | 36905 | 668 |
| Kansas | 340 × 650 | 400–2500 nm | 330 | 7 | 41953 | 728 |

The Chikusei dataset is adopted as the source domain and is widely used as a benchmark dataset in few-shot and cross-domain HSI studies. It was collected over Chikusei City, Ibaraki Prefecture, Japan, using the Hyperspec-VNIR-C visible/near-infrared hyperspectral sensor. The image has spatial dimensions of 2517 × 2335 pixels with 128 spectral bands covering the wavelength range of 363–1018 nm. A total of 19 land-cover classes are provided, with abundant labeled samples and high category diversity, making this dataset particularly suitable for learning transferable base representations. The corresponding pseudo-color image and ground-truth map are shown in Fig. 7.

The Indian Pines (IP) dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over northwestern Indiana, USA, and is one of the most commonly used benchmark datasets for cross-domain HSI classification. The original data consist of 220 spectral bands spanning 400–2500 nm, of which 200 bands are retained after removing those affected by water-vapor absorption. The image has spatial dimensions of 145 × 145 pixels. The pseudo-color image and ground-truth map are presented in Fig. 8.

The Houston (HT) dataset was collected by the ITRES CASI-1500 sensor and is a representative urban hyperspectral scene released as part of the IEEE GRSS 2013 Data Fusion Contest. The image has spatial dimensions of 349 × 1905 pixels with 144 spectral bands in the wavelength range of 364–1046 nm and a spatial resolution of approximately 2.5 m. It includes 15 land-cover categories. The pseudo-color image and ground-truth map are shown in Fig. 9.

The WHU-Hi-HongHu (HH) dataset is a typical wetland–farmland hyperspectral scene published by Wuhan University. The original data contain 270 spectral bands ranging from 400 to 1000 nm. It includes 18 land-cover categories. Notably, since the HH dataset contains a large number of samples, following [39], we randomly select 10% of the samples from each category. The image has spatial dimensions of 940 × 475 pixels and was collected using Wuhan University's airborne hyperspectral imaging system.

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The pseudo-color image and ground-truth map are shown in Fig. 10.

The Kansas AHSI dataset was acquired by the Advanced Hyperspectral Imager (AHSI) covering the visible–shortwave infrared (VIS–SWIR) range, developed by the Shanghai Institute of Technical Physics of the Chinese Academy of Sciences. As a key payload of the GF-5 satellite, this instrument acquires data with a spatial size of 340 × 650 pixels. It covers the spectral range of 400–2500 nm with 330 spectral bands in the solar-reflective region. The pseudo-color image and corresponding ground-truth map are shown in Fig. 11.

### B. Experimental Settings

To comprehensively evaluate the effectiveness of the proposed SpikeFSL framework, we conduct systematic comparisons with several representative methods, including two traditional supervised learning approaches, namely SVM [28] and SSRN [29], one classical single-domain SNN-based model for HSI classification, i.e., HSI-SNN [19], and a set of representative FSL methods for cross-domain HSI classification, including DCFSL [11], CMFSL [30], Gia-CFSL [31], HyMuT [36], MRLF [35], SGPL [38], and TEFSL [39]. These comparison methods collectively cover metric learning, graph-based learning, semantic-guided learning, and transductive inference paradigms, providing a comprehensive evaluation baseline.

Under the cross-domain FSL setting, the training process follows an episodic strategy. Specifically, for each class in the source domain, 200 labeled samples are randomly selected to learn transferable representations, while in the target domain, only a very limited number of labeled samples per class are provided for training, with all remaining samples reserved for testing. This setting is consistent with existing cross-domain HSI classification studies and is designed to simulate realistic cross-domain scenarios.

For the proposed SpikeFSL, the spatial neighborhood size of each pixel is fixed at 9 × 9. The number of spectral channels is set to 100, and the dimensionality of the final feature representations is set to 160. The proposed framework is optimized using the Adam optimizer, with the learning rate set to $3 \times 10^{-4}$ for the IP, HT, and Kansas datasets and $1 \times 10^{-4}$ for the HH dataset. Each experiment is independently repeated ten times to alleviate randomness. For comparison methods, the number of training iterations is fixed at 10 000 following common practice in previous cross-domain FSL studies. For SpikeFSL, we use 1000 training iterations, since the proposed method already achieves effective target-domain generalization under this setting. The final results are reported in the form of mean ± standard deviation.

Performance is evaluated using widely adopted metrics in cross-domain HSI classification, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient, providing a comprehensive assessment of overall performance, class-balanced accuracy, and classification agreement. All experiments are conducted on a workstation equipped with an Intel Xeon Silver 4130 CPU, an NVIDIA GeForce RTX 4090 GPU, and 32 GB of DDR4 memory, and all methods are implemented using the PyTorch framework.

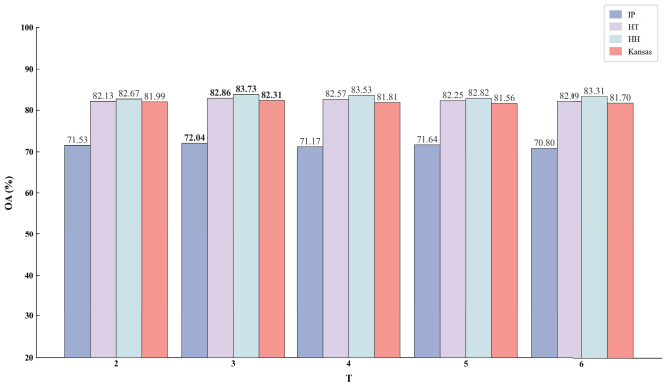



Fig. 12. OA (%) on four target-domain datasets with respect to different time-step $T$ values.

TABLE II
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT $\lambda_{FA}$ VALUES

| $\lambda_{FA}$ | 0.1 | 0.3 | 0.5 | 0.7 | 0.9 |
|---|---|---|---|---|---|
| IP | **72.04** | 71.62 | 71.70 | 71.63 | 71.59 |
| HT | 82.16 | 81.97 | 82.11 | 82.41 | **82.86** |
| HH | 83.38 | 83.23 | 83.28 | **83.62** | 83.43 |
| Kansas | 81.43 | 81.60 | 81.63 | **82.02** | 81.92 |

TABLE III
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT $\lambda_{HNC}$ VALUES

| $\lambda_{HNC}$ | 0.5 | 1.5 | 3.0 | 4.5 | 5.5 |
|---|---|---|---|---|---|
| IP | 71.44 | 71.41 | **72.04** | 71.96 | 71.75 |
| HT | 82.14 | 82.31 | **82.86** | 82.47 | 82.42 |
| HH | 83.30 | 83.36 | **83.62** | 82.34 | 82.48 |
| Kansas | 81.61 | **82.11** | 82.02 | 81.91 | 81.84 |

### C. Hyperparameter Tuning

To ensure stable and robust cross-domain classification, systematic tuning is performed on six key hyperparameters in SpikeFSL: the weight coefficient $\lambda_{FA}$ for the feature alignment loss $L_{FA}$, the weight coefficient $\lambda_{HNC}$ for the hard-negative contrastive loss $L_{HNC}$, the base hyperparameter $\beta_0$ and temperature parameter $\tau$ that control the strength of prototype refinement, and the membrane potential firing threshold $v_{th}$ and time step $T$ of the SNN. The candidate values are set as follows: $\lambda_{FA} \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, $\lambda_{HNC} \in \{0.5, 1.5, 3.0, 4.5, 5.5\}$, $\beta_0 \in \{0.5, 0.8, 1.0, 1.2, 1.5\}$, $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, $v_{th} \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, and $T \in \{2, 3, 4, 5, 6\}$.

To reduce the computational burden of joint hyperparameter search, a single-factor tuning strategy is employed, where one parameter is varied while the others are fixed to their default values. After determining the optimal value for a given parameter, the next parameter is tuned sequentially. All configurations are independently evaluated on the four target-domain datasets using OA as the evaluation metric. The quantitative results and performance trends are reported in Tables II–VI and Fig. 12.

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TABLE IV
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT $\beta_0$ VALUES

| $\beta_0$ | 0.5 | 0.8 | 1.0 | 1.2 | 1.5 |
|---|---|---|---|---|---|
| IP | 71.16 | 72.00 | **72.04** | 71.61 | 71.45 |
| HT | 82.47 | 81.51 | **82.86** | 82.62 | 82.44 |
| HH | **83.67** | 83.54 | 83.62 | 83.07 | 83.13 |
| Kansas | 81.69 | 81.65 | 82.11 | 81.98 | **82.22** |

TABLE V
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT $\tau$ VALUES

| $\tau$ | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 |
|---|---|---|---|---|---|
| IP | 71.69 | 71.56 | **72.04** | 71.24 | 71.58 |
| HT | 81.62 | 82.17 | **82.86** | 82.76 | 82.16 |
| HH | **83.73** | 83.09 | 83.67 | 83.52 | 83.64 |
| Kansas | **82.31** | 81.98 | 82.22 | 82.00 | 81.72 |

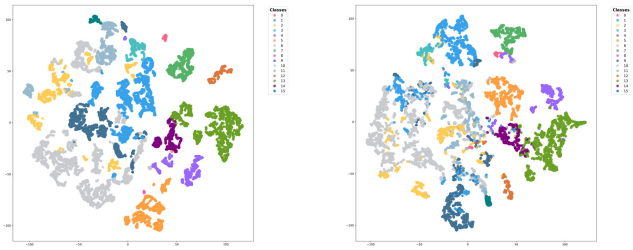
TABLE VI
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT $v_{\text{th}}$ VALUES

| $v_{\text{th}}$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 |
|---|---|---|---|---|---|
| IP | **72.04** | 71.19 | 70.79 | 70.52 | 70.68 |
| HT | **82.86** | 81.84 | 81.06 | 80.81 | 80.86 |
| HH | **83.73** | 82.87 | 82.34 | 82.45 | 82.60 |
| Kansas | **82.31** | 80.18 | 79.80 | 80.05 | 79.80 |

The optimal hyperparameter configurations vary across datasets, reflecting differences in their spectral and structural characteristics. A low $\lambda_{\text{FA}}$ of 0.1 is optimal for IP due to its moderate spectral mixing and limited domain shift, where excessive alignment may hinder class separability. In contrast, HT requires a high $\lambda_{\text{FA}}$ of 0.9 to enforce stronger cross-paradigm consistency, which may be attributed to its complex urban land cover and pronounced domain discrepancies. Both HH and Kansas achieve the best performance with a moderate $\lambda_{\text{FA}}$ of 0.7, balancing alignment strength and feature flexibility under moderate domain variations. For $\lambda_{\text{HNC}}$, IP, HT, and HH all select 3.0, indicating that enhancing prototype discriminability is important for handling spectrally similar classes in these datasets. Kansas adopts a lower $\lambda_{\text{HNC}}$ of 1.5, likely because its land-cover categories are more separable, reducing the need for strong margin enforcement. The base hyperparameter $\beta_0$ and temperature $\tau$ jointly modulate the focus on hard negatives. IP and HT employ $\beta_0 = 1.0$ and $\tau = 0.5$, offering a balanced emphasis on difficult samples and moderate similarity sharpness. HH uses $\beta_0 = 0.5$ and $\tau = 0.3$. The lower $\beta_0$ softens the hard-negative weighting, while the smaller $\tau$ concentrates the similarity distribution, which suits its clearer class boundaries. Kansas combines a higher $\beta_0 = 1.5$ with $\tau = 0.3$ to place stronger emphasis on hard negatives, compensating for potential feature ambiguity. Finally, all datasets share the same optimal $v_{\text{th}} = 0.1$ and $T = 3$. A lower firing threshold allows more informative signals to be preserved during spike generation, thereby improving feature representation quality. For static HSI data, the introduced temporal dimension is not a physical temporal attribute but a computational unfolding required by SNNs. Three time steps achieve a favorable tradeoff between accuracy and computational cost, possibly because the SSTGSE TA helps each step focus on complementary spatial regions, whereas fewer steps limit diversity and more steps can introduce redundancy that weakens discriminability.

TABLE VII
OA (%) COMPARISON OF SNN AND CNN BRANCH SYNERGY IN LSCFEN UNDER THE ABLATION STUDY

| Case | SSTGSE | LSCFEN | IFPR | IP | HT | HH | Kansas |
|---|---|---|---|---|---|---|---|
| 1 | | | | 60.28 | 61.74 | 63.25 | 71.88 |
| 2 | ✓ | | | 67.97 | 80.44 | 79.15 | 80.95 |
| 3 | ✓ | ✓ | | 70.83 | 81.13 | 82.89 | 81.50 |
| 4 | ✓ | ✓ | ✓ | **72.04** | **82.86** | **83.73** | **82.31** |



(a) (b)

Fig. 13. t-SNE visualizations of features extracted from the IP dataset by (a) SNN branch only and (b) 1D-CNN branch only.

## D. Ablation and Comparative Study

To quantitatively assess the contribution of each key component in the proposed SpikeFSL framework, an ablation study is conducted using OA as the evaluation metric. Four network variants are constructed by progressively introducing the proposed modules. Case 1 serves as the baseline model, which employs only the spike convolution block for feature extraction. Case 2 augments the baseline with the SSTGSE module. Case 3 further incorporates the LSCFEN, i.e., the 1-D spectral convolution block, enabling cross-paradigm feature learning. Case 4 introduces the IFPR strategy, forming the complete SpikeFSL framework. The 5-shot classification results on the IP, HT, HH, and Kansas datasets are summarized in Table VII.

As shown in Table VII, the baseline model achieves relatively low performance across all four datasets. After introducing SSTGSE, the classification accuracy improves substantially on all datasets, with the most significant gain observed on the HT dataset. This demonstrates that SSTGSE effectively transforms raw HSI pixels into more informative spike-compatible representations and provides better inputs for subsequent spiking-based feature extraction.

Building upon SSTGSE, the incorporation of the 1-D Spectral Convolution Block into LSCFEN leads to further performance improvements on all datasets. To further verify that the performance gain stems from the synergy between

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TABLE VIII
OA (%) COMPARISON OF SNN AND CNN BRANCH SYNERGY IN LSCFEN UNDER THE ABLATION STUDY

| SNN branch | 1D-CNN branch | IP | HT | HH | Kansas | Average |
|---|---|---|---|---|---|---|
| | ✓ | 66.38 | 79.16 | 81.74 | 78.10 | 76.35 |
| ✓ | | 67.97 | 80.44 | 79.15 | 80.95 | 77.13 |
| ✓ | ✓ | **70.83** | **81.13** | **82.89** | **81.50** | **79.09** |

TABLE IX
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT FEATURE FUSION STRATEGIES

| | Adaptive Weighting | Cross-attention | Average fusion |
|---|---|---|---|
| IP | 71.57 | 70.67 | **72.04** |
| HT | 82.78 | 80.75 | **82.86** |
| HH | 82.40 | 81.97 | **83.73** |
| Kansas | 82.15 | 80.45 | **82.31** |

TABLE X
OA (%) ON FOUR TARGET-DOMAIN DATASETS UNDER DIFFERENT ALIGNMENT METHODS

| | MMD | LMMD | CKA |
|---|---|---|---|
| IP | 70.92 | 70.98 | **72.04** |
| HT | 80.97 | 80.42 | **82.86** |
| HH | 83.60 | 83.42 | **83.73** |
| Kansas | 81.19 | 81.24 | **82.31** |

SNN and CNN, additional ablation experiments are conducted on LSCFEN with three configurations: the 1D-CNN branch only, the SNN branch only, and the dual-branch fusion. The results are summarized in Table VIII. The SNN-only branch achieves a slightly higher average OA than the 1D-CNN-only branch, suggesting its advantage in learning more transferable representations under the cross-domain setting. More importantly, the dual-branch model attains the highest OA across all datasets, which further confirms that the complementary integration of dense spectral modeling and sparse spiking features enhances feature expressiveness. Furthermore, Fig. 13 presents the t-SNE visualizations of features extracted by the two single-branch models on the IP dataset, where the SNN branch generates more compact intraclass clusters and clearer interclass boundaries. Finally, the complete SpikeFSL framework with the IFPR module achieves the best performance on all datasets.

We further compare feature fusion strategies within LSCFEN, and the results are summarized in Table IX. Average fusion consistently achieves better performance than adaptive weighting and cross-attention across all datasets. Although the latter two introduce additional learnable parameters, their lower accuracy may be attributed to overfitting in few-shot scenarios, where complex fusion mechanisms struggle to generalize. Average fusion effectively balances the complementary strengths of sparse SNN features and dense CNN features without extra complexity, offering a more robust and generalizable fusion scheme.

We also compare CKA with MMD and LMMD for cross-paradigm alignment in IFPR, as reported in Table X. CKA consistently achieves the best accuracy on all datasets. This result supports the motivation that, for heterogeneous SNN–CNN representations, structure alignment is more appropriate than direct distribution alignment. MMD and LMMD mainly reduce discrepancy by matching feature distributions, which may overconstrain sparse spike-driven features and dense continuous features into a similar distribution form. In contrast, CKA aligns the intersample similarity structure induced by the two branches, thereby preserving their branch-specific encoding characteristics while promoting semantic consistency across paradigms. This property makes CKA more suitable for heterogeneous SNN–CNN feature fusion under domain shift.

### *E. Comparison With Other Methods*

This section evaluates the performance of the proposed SpikeFSL framework on cross-domain HSI classification tasks. We compare SpikeFSL with conventional supervised learning methods, namely SVM and SSRN, a single-domain SNN model, HSI-SNN, as well as several state-of-the-art cross-domain FSL approaches, including DCFSL, CMFSL, Gia-CFSL, HyMuT, MRLF, SGPL, and TEFSL. Following common practice in cross-domain FSL evaluation, a 5-shot setting is employed, where five labeled samples per class from the target domain are randomly selected as support samples. Each experiment is repeated ten times with different random seeds, and the mean and standard deviation are reported.

Table XI presents the 5-shot classification results on the IP dataset. The results show that traditional supervised methods yield relatively limited performance due to their reliance on sufficient labeled data and lack of cross-domain adaptation capability. The single-domain HSI-SNN model also performs poorly under the cross-domain FSL setting, as it is not designed to handle domain shifts or adapt with very limited support samples. In contrast, FSL-based methods achieve substantially higher accuracy by leveraging episodic training and transferable knowledge learned from the source domain. Among them, the proposed SpikeFSL framework achieves the best performance, reaching 72.04% OA, 82.88% AA, and a Kappa coefficient of 68.50, clearly outperforming all competing approaches.

Similar trends can be observed in Tables XII–XIV, which report results on the HT, HH, and Kansas datasets, respectively. SpikeFSL consistently achieves the highest OA, AA, and Kappa values across all three datasets, demonstrating strong robustness to variations in sensors, spatial resolutions, and land-cover distributions.

To further illustrate the qualitative differences among methods, classification maps on the IP, HT, HH, and Kansas datasets are shown in Figs. 14–17. Supervised learning methods and the single-domain SNN model exhibit noticeable misclassifications, noisy regions, and blurred class boundaries. In contrast, FSL-based methods generate more coherent classification maps. Among them, SpikeFSL produces the most visually consistent results, particularly in boundary regions

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TABLE XI

CLASSIFICATION PERFORMANCE (%) OF VARIOUS ALGORITHMS ON IP UNDER THE 5-SHOT SETTING

| Class | SVM | SSRN | HSI-SNN | DCFSL | CMFSL | Gia-CFSL | HyMuT | MRLF | SGPL | TEFSL | SpikeFSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 72.20 | 18.38 | 83.41 | 92.20 | 98.54 | 90.73 | 96.34 | 96.34 | 30.42 | 90.00 | **99.51** |
| 2 | 34.27 | 64.79 | 29.31 | 39.52 | 60.94 | 44.05 | 53.01 | 48.51 | **75.19** | 47.96 | 58.33 |
| 3 | 39.12 | 27.65 | 38.88 | 49.72 | 61.78 | 49.59 | 61.38 | 55.72 | 48.61 | 59.60 | **64.91** |
| 4 | 50.34 | 26.97 | 60.34 | 80.22 | 85.60 | 79.09 | 83.84 | 86.94 | 45.97 | 79.96 | **87.93** |
| 5 | 69.75 | 80.76 | 66.95 | 73.31 | 78.74 | 72.47 | 77.22 | 78.95 | **81.76** | 76.57 | 80.73 |
| 6 | 66.36 | 86.87 | 78.43 | 84.92 | 93.01 | 83.50 | 86.03 | 93.01 | 90.50 | 87.82 | **93.77** |
| 7 | 89.13 | 32.24 | 99.57 | 96.96 | **100.00** | 96.52 | **100.00** | 99.13 | 47.27 | 98.26 | **100.00** |
| 8 | 68.73 | **100.00** | 76.34 | 82.24 | 85.71 | 87.08 | 90.63 | 84.08 | 78.72 | 88.16 | 86.36 |
| 9 | 86.67 | 57.69 | **100.00** | **100.00** | **100.00** | 98.67 | 99.33 | 99.67 | 16.41 | **100.00** | **100.00** |
| 10 | 37.49 | 59.69 | 48.35 | 58.67 | 58.66 | 59.02 | 66.23 | **67.79** | 47.29 | 62.22 | 66.47 |
| 11 | 33.96 | 70.87 | 43.44 | 58.70 | 55.04 | 62.65 | 59.22 | 66.16 | **78.72** | 58.95 | 62.27 |
| 12 | 31.43 | 45.00 | 27.67 | 46.26 | **62.47** | 45.80 | 53.08 | 53.03 | 47.66 | 44.23 | 56.55 |
| 13 | 86.50 | 88.29 | 87.50 | 97.85 | 96.55 | 98.90 | 98.20 | **99.65** | 57.87 | 98.90 | 98.60 |
| 14 | 62.93 | **97.18** | 87.50 | 85.86 | 76.28 | 78.60 | 81.44 | 87.83 | 79.37 | 85.78 | 86.92 |
| 15 | 28.08 | 36.64 | 61.31 | 74.17 | 80.39 | 69.37 | 78.64 | 72.70 | 49.58 | 71.34 | **84.59** |
| 16 | 90.91 | 60.98 | 93.98 | 99.55 | **99.66** | 98.64 | 99.09 | 99.09 | 42.05 | 98.86 | 99.20 |
| **OA** | $45.85 \pm 2.44$ | $61.36 \pm 0.49$ | $53.97 \pm 3.67$ | $64.06 \pm 3.54$ | $68.26 \pm 4.12$ | $64.62 \pm 3.78$ | $68.51 \pm 3.49$ | $70.18 \pm 2.05$ | $64.68 \pm 0.95$ | $66.84 \pm 2.64$ | $\mathbf{72.04 \pm 2.82}$ |
| **AA** | $59.24 \pm 1.36$ | $59.75 \pm 0.20$ | $67.69 \pm 2.95$ | $76.26 \pm 1.96$ | $80.84 \pm 1.44$ | $75.92 \pm 2.24$ | $80.23 \pm 1.76$ | $80.47 \pm 1.89$ | $58.10 \pm 0.89$ | $78.04 \pm 1.03$ | $\mathbf{82.88 \pm 1.39}$ |
| **Kappa** | $39.68 \pm 2.48$ | $56.91 \pm 0.48$ | $48.87 \pm 3.78$ | $59.58 \pm 3.83$ | $64.37 \pm 4.26$ | $60.04 \pm 3.97$ | $64.56 \pm 3.69$ | $66.36 \pm 3.69$ | $60.61 \pm 0.94$ | $62.64 \pm 2.73$ | $\mathbf{68.50 \pm 3.03}$ |

TABLE XII

CLASSIFICATION PERFORMANCE (%) OF VARIOUS ALGORITHMS ON HT UNDER THE 5-SHOT SETTING

| Class | SVM | SSRN | HSI-SNN | DCFSL | CMFSL | Gia-CFSL | HyMuT | MRLF | SGPL | TEFSL | SpikeFSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 81.21 | 83.76 | 75.94 | 88.48 | 88.39 | 88.84 | 89.55 | 88.94 | 85.37 | **91.12** | 88.72 |
| 2 | 45.80 | 68.11 | 72.60 | 86.27 | 91.47 | 89.05 | 90.56 | 89.05 | 72.11 | 91.49 | **94.14** |
| 3 | **98.62** | 49.42 | 95.04 | 95.65 | 97.34 | 94.94 | 96.10 | 97.66 | 92.50 | 96.33 | 97.33 |
| 4 | 88.88 | 92.78 | 75.84 | 95.76 | **98.26** | 93.97 | 96.25 | 97.15 | 74.12 | 93.76 | 96.87 |
| 5 | 74.08 | 90.93 | 93.81 | 88.94 | 91.18 | 97.24 | 97.90 | 91.18 | 96.54 | 97.75 | **98.71** |
| 6 | 82.30 | **87.66** | 83.31 | **87.66** | 84.25 | 81.47 | 81.31 | 81.31 | 87.18 | 76.44 | 84.03 |
| 7 | 12.87 | 75.52 | 41.62 | 76.10 | **89.50** | 73.75 | 78.61 | 83.58 | 78.61 | 76.77 | 76.03 |
| 8 | 14.03 | 58.80 | 27.72 | 58.19 | 54.78 | 49.63 | 51.49 | 52.75 | **72.33** | 48.64 | 51.48 |
| 9 | 75.92 | 67.11 | 48.23 | 77.73 | 72.65 | 65.12 | 69.93 | 75.90 | **79.61** | 64.59 | 77.56 |
| 10 | 5.29 | 55.52 | 52.33 | 52.40 | 63.24 | 68.63 | 71.35 | 68.38 | 73.46 | **75.21** | 73.29 |
| 11 | 28.65 | 71.07 | 55.83 | 76.93 | 74.96 | 65.46 | 65.46 | 76.32 | **81.97** | 74.72 | 76.54 |
| 12 | 5.39 | 38.76 | 39.87 | 38.88 | 59.36 | 65.92 | 62.44 | 62.77 | **74.39** | 69.62 | 69.53 |
| 13 | 10.35 | 80.62 | 74.03 | 80.69 | 93.84 | 90.13 | 90.13 | 90.62 | 37.01 | **95.86** | 91.88 |
| 14 | 95.01 | **99.08** | 96.90 | **99.08** | 82.86 | 91.96 | 93.50 | 92.32 | 65.30 | 97.71 | 98.27 |
| 15 | 92.83 | 98.17 | 98.17 | 98.17 | 98.84 | **99.63** | 98.12 | 98.12 | 85.54 | 98.37 | 98.85 |
| **OA** | $45.55 \pm 3.11$ | $72.02 \pm 3.55$ | $63.93 \pm 5.60$ | $79.61 \pm 1.79$ | $81.25 \pm 1.87$ | $78.39 \pm 1.72$ | $80.66 \pm 3.11$ | $79.79 \pm 1.77$ | $74.93 \pm 1.02$ | $81.14 \pm 2.57$ | $\mathbf{82.86 \pm 2.28}$ |
| **AA** | $54.08 \pm 2.55$ | $74.46 \pm 3.65$ | $68.75 \pm 4.95$ | $82.60 \pm 2.70$ | $83.44 \pm 1.65$ | $80.53 \pm 1.47$ | $82.60 \pm 2.70$ | $81.79 \pm 1.58$ | $77.07 \pm 1.26$ | $83.22 \pm 2.05$ | $\mathbf{84.88 \pm 1.93}$ |
| **Kappa** | $44.80 \pm 3.27$ | $69.76 \pm 3.85$ | $61.18 \pm 5.99$ | $77.97 \pm 2.02$ | $79.73 \pm 2.02$ | $76.66 \pm 1.85$ | $79.10 \pm 3.36$ | $75.90 \pm 1.91$ | $79.63 \pm 1.09$ | $79.10 \pm 2.77$ | $\mathbf{81.48 \pm 2.47}$ |

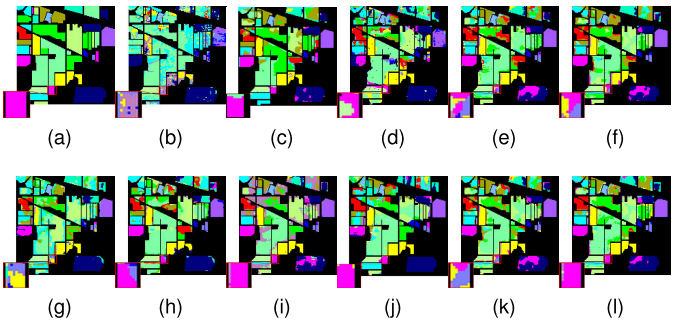



Fig. 14. Classification maps on IP under the 5-shot setting. (a) Ground truth, (b) SVM, (c) SSRN, (d) HSI-SNN, (e) DCFSL, (f) CMFSL, (g) Gia-CFSL, (h) HyMuT, (i) MRLF, (j) SGPL, (k) TEFSL, and (l) SpikeFSL.

and small-object areas, reflecting its ability to learn robust and discriminative cross-domain representations.

To further analyze the learned representations, t-SNE visualizations are provided to compare the feature distributions produced by SGPL, TEFSL, and SpikeFSL on the four target datasets, as shown in Fig. 18. Compared with SGPL and TEFSL, SpikeFSL yields more compact intraclass clusters and clearer interclass separation. For example, on the IP dataset, several spectrally similar classes become more distinguishable, while on the HT dataset, urban categories show improved clustering compactness. These observations further confirm the effectiveness of SpikeFSL in learning structured and discriminative cross-domain feature representations.

Overall, the superior performance of SpikeFSL can be attributed to the following factors.

1) Conventional supervised learning methods and single-domain SNN models lack the ability to learn cross-domain transferable representations, leading to poor generalization under cross-domain FSL settings.
2) Compared with current FSL methods, SpikeFSL introduces spike-driven representation learning into the cross-domain FSL framework, effectively leveraging

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TABLE XIII
CLASSIFICATION PERFORMANCE (%) OF VARIOUS ALGORITHMS ON HH UNDER THE 5-SHOT SETTING

| Class | SVM | SSRN | HSI-SNN | DCFSL | CMFSL | Gia-CFSL | HyMuT | MRLF | SGPL | TEFSL | SpikeFSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 52.98 | 85.60 | 76.08 | 80.06 | 71.92 | 77.76 | 81.30 | 83.74 | 87.95 | 84.80 | **89.59** |
| 2 | 38.88 | 62.83 | 52.34 | 76.82 | 67.77 | 68.18 | 72.83 | 76.07 | 60.13 | 71.85 | **79.31** |
| 3 | 80.32 | 86.80 | 83.92 | 91.80 | 79.44 | 88.96 | 90.24 | 87.14 | 84.40 | **91.97** | 88.20 |
| 4 | 52.91 | 88.87 | 70.22 | 86.28 | 80.14 | 85.27 | 86.74 | 87.41 | **99.06** | 88.15 | 91.82 |
| 5 | 10.66 | 53.91 | 58.81 | 70.70 | 76.49 | 65.63 | 72.01 | 69.48 | 43.52 | 69.11 | **79.61** |
| 6 | 39.36 | 74.94 | 74.99 | 86.89 | 82.02 | 82.86 | 87.57 | 87.02 | **90.52** | 87.62 | 88.03 |
| 7 | 30.46 | 65.99 | 37.23 | 56.77 | 52.36 | 52.67 | 58.39 | 60.64 | 65.26 | 61.92 | **68.69** |
| 8 | 6.43 | 23.12 | 27.52 | 36.37 | 48.93 | 26.78 | 40.95 | 33.80 | 41.93 | 38.50 | **52.90** |
| 9 | 34.92 | 95.40 | 90.62 | 93.60 | 86.87 | 93.36 | 94.26 | 93.92 | 93.95 | **96.58** | 95.88 |
| 10 | 12.28 | 64.83 | 38.25 | 59.46 | 54.85 | 56.92 | 62.54 | 60.48 | **74.23** | 62.71 | 66.88 |
| 11 | 9.11 | 64.58 | 31.52 | 51.72 | 59.73 | 53.73 | 58.48 | 56.09 | 47.70 | 56.69 | **68.07** |
| 12 | 9.06 | **73.57** | 59.92 | 60.99 | 44.55 | 62.29 | 60.37 | 59.18 | 61.58 | 58.79 | 58.54 |
| 13 | 16.17 | 41.39 | 43.19 | 45.97 | 47.36 | 51.48 | 51.86 | 50.47 | **85.65** | 54.00 | 54.53 |
| 14 | 23.83 | **80.96** | 68.00 | 76.60 | 60.86 | 77.55 | 72.70 | 77.71 | 39.41 | 78.00 | 78.15 |
| 15 | 3.04 | **94.53** | 81.89 | 88.00 | 93.89 | 86.95 | 84.95 | 86.21 | 12.11 | 88.63 | 90.42 |
| 16 | 51.65 | 78.81 | 72.57 | 60.82 | 82.12 | 67.39 | 70.50 | 73.29 | **94.94** | 75.30 | 80.90 |
| 17 | 29.82 | 88.55 | 49.70 | 79.76 | 80.98 | 78.85 | 76.32 | 82.80 | **90.42** | 76.08 | 81.69 |
| 18 | 7.32 | 87.75 | 70.51 | 81.14 | **91.39** | 77.97 | 74.84 | 82.12 | 36.96 | 83.69 | 91.27 |
| **OA** | $38.85 \pm 8.60$ | $80.10 \pm 3.89$ | $64.62 \pm 10.97$ | $78.06 \pm 3.07$ | $73.36 \pm 4.07$ | $76.63 \pm 3.01$ | $79.28 \pm 3.09$ | $79.34 \pm 2.53$ | $82.88 \pm 1.17$ | $80.54 \pm 2.52$ | $\mathbf{83.73 \pm 2.04}$ |
| **AA** | $28.34 \pm 0.97$ | $73.30 \pm 3.42$ | $59.09 \pm 6.67$ | $71.32 \pm 1.96$ | $70.09 \pm 2.80$ | $69.21 \pm 2.24$ | $72.53 \pm 2.66$ | $72.64 \pm 1.89$ | $67.19 \pm 0.73$ | $73.56 \pm 1.73$ | $\mathbf{78.03 \pm 1.18}$ |
| **Kappa** | $31.31 \pm 6.76$ | $74.94 \pm 4.50$ | $57.65 \pm 11.41$ | $72.55 \pm 3.41$ | $67.10 \pm 4.38$ | $70.82 \pm 3.47$ | $74.05 \pm 3.56$ | $74.02 \pm 2.86$ | $78.27 \pm 1.28$ | $75.53 \pm 2.88$ | $\mathbf{79.36 \pm 2.41}$ |

TABLE XIV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS ALGORITHMS ON KANSAS UNDER THE 5-SHOT SETTING

| Class | SVM | SSRN | HSI-SNN | DCFSL | CMFSL | Gia-CFSL | HyMuT | MRLF | SGPL | TEFSL | SpikeFSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 82.51 | 77.26 | 71.27 | 81.69 | 47.72 | 84.59 | 83.77 | 85.64 | 80.58 | 87.06 | **89.49** |
| 2 | **97.21** | 79.97 | 80.23 | 91.26 | 80.87 | 95.44 | 91.84 | 96.05 | 92.25 | 91.77 | 95.31 |
| 3 | 59.23 | 77.68 | 60.88 | 74.85 | 45.63 | 82.26 | 77.22 | 80.86 | 75.07 | 73.66 | **83.80** |
| 4 | 76.91 | 41.31 | 61.65 | 72.01 | 72.32 | 80.54 | 75.07 | 77.43 | **94.37** | 77.77 | 77.71 |
| 5 | 71.66 | 69.58 | 44.35 | 84.80 | 64.15 | 89.45 | 88.57 | 90.14 | 74.65 | 89.10 | **93.42** |
| 6 | 51.97 | **95.33** | 60.85 | 78.96 | 64.52 | 85.52 | 80.14 | 78.34 | 58.21 | 81.72 | 82.25 |
| 7 | 42.73 | 38.59 | 14.50 | 49.18 | 36.34 | 51.13 | 50.65 | 36.30 | 38.91 | **58.18** | 48.73 |
| **OA** | $71.60 \pm 0.13$ | $62.29 \pm 1.46$ | $56.60 \pm 5.92$ | $76.33 \pm 4.31$ | $61.55 \pm 5.70$ | $81.91 \pm 2.92$ | $78.76 \pm 4.63$ | $79.30 \pm 4.50$ | $78.66 \pm 1.44$ | $80.53 \pm 3.61$ | $\mathbf{82.31 \pm 2.52}$ |
| **AA** | $68.89 \pm 0.16$ | $68.53 \pm 0.93$ | $56.25 \pm 4.53$ | $76.11 \pm 3.97$ | $58.79 \pm 4.91$ | $81.28 \pm 2.65$ | $78.18 \pm 4.51$ | $77.82 \pm 4.63$ | $73.43 \pm 2.05$ | $79.89 \pm 2.82$ | $\mathbf{81.57 \pm 1.99}$ |
| **Kappa** | $65.98 \pm 0.18$ | $56.08 \pm 1.53$ | $48.37 \pm 6.59$ | $71.65 \pm 5.03$ | $53.95 \pm 6.46$ | $78.17 \pm 3.43$ | $74.47 \pm 5.42$ | $74.97 \pm 5.42$ | $74.10 \pm 1.76$ | $76.56 \pm 4.20$ | $\mathbf{78.68 \pm 2.93}$ |

the sparse and event-driven characteristics of SNNs to achieve more robust representations with reduced sensitivity to noise and redundant information.

3) The proposed LSCFEN jointly exploits spiking-based spectral–spatial–temporal features and fine-grained spectral representations, yielding lightweight yet discriminative features for cross-domain matching.
4) The IFPR strategy aligns semantic geometry between SNN and CNN via CKA, facilitates complementary fusion of their respective strengths, and focuses on hard-negative samples for prototype refinement, thereby enhancing cross-domain robustness.

### *F. Impact of the Number of Labeled Samples on Performance*

To further evaluate the robustness of the proposed SpikeFSL framework with respect to the number of labeled samples and its cross-domain generalization capability, experiments are conducted under different labeled-sample settings on the four target-domain datasets. Specifically, for FSL-based methods, 1, 2, 3, 4, and 5 labeled samples per class are randomly selected from each target domain for cross-domain classification. For supervised learning methods that do not employ episodic cross-domain training, the same numbers of labeled samples per class are used for model training to ensure a fair comparison.

As illustrated in Fig. 19, the OA of most compared methods improves as the number of labeled samples increases. Across different labeled-sample settings, SpikeFSL achieves the highest OA in the majority of cases on the four target datasets, and its advantage becomes more pronounced as more labeled samples are available. These results indicate that SpikeFSL maintains stable and competitive performance even under extremely limited supervision.

It is worth noting that the HSI-SNN and SGPL methods have inherent limitations in their original implementations and training protocols, which make them unsuitable for reliable evaluation under the 1-shot setting. Therefore, results for these two methods under the 1-shot configuration are not reported in Fig. 19 to ensure a fair and consistent comparison.

### *G. Impact of the Number of Training Iterations on Performance*

To further clarify the influence of training iterations on cross-domain generalization and explain the iteration settings adopted in this work, we evaluate the OA of different FSL methods under 1000 and 10 000 training iterations. As shown

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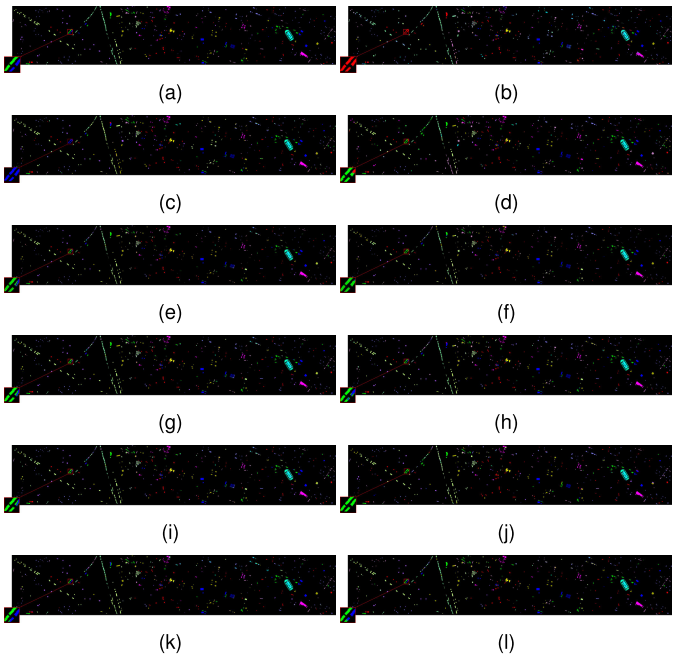



Fig. 15. Classification maps on HT under the 5-shot setting. (a) Ground truth, (b) SVM, (c) SSRN, (d) HSI-SNN, (e) DCFSL, (f) CMFSL, (g) Gia-CFSL, (h) HyMuT, (i) MRLF, (j) SGPL, (k) TEFSL, and (l) SpikeFSL.

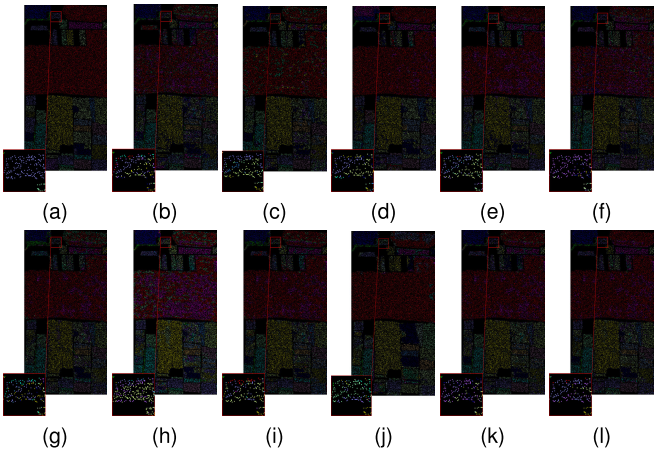



Fig. 16. Classification maps on HH under the 5-shot setting. (a) Ground truth, (b) SVM, (c) SSRN, (d) HSI-SNN, (e) DCFSL, (f) CMFSL, (g) Gia-CFSL, (h) HyMuT, (i) MRLF, (j) SGPL, (k) TEFSL, and (l) SpikeFSL.

in Table XV, most competing methods still exhibit evident performance improvements when the number of iterations increases from 1000 to 10 000, indicating that they generally require more training iterations to achieve effective target-domain generalization. For example, the OA gain of SGPL reaches 12.40% on the HT dataset, while HyMuT and CMFSL obtain improvements of 11.94% and 9.88% on the IP and HH datasets, respectively. In contrast, SpikeFSL shows only marginal performance variations across all four target-domain datasets, with OA improvements of 0.58%, 1.83%, 0.10%, and 0.39% on IP, HT, HH, and Kansas, respectively. Moreover, even after being trained for 10 000 iterations, the best results of the competing methods are still lower than those achieved by SpikeFSL with only 1000 iterations. These results indicate that

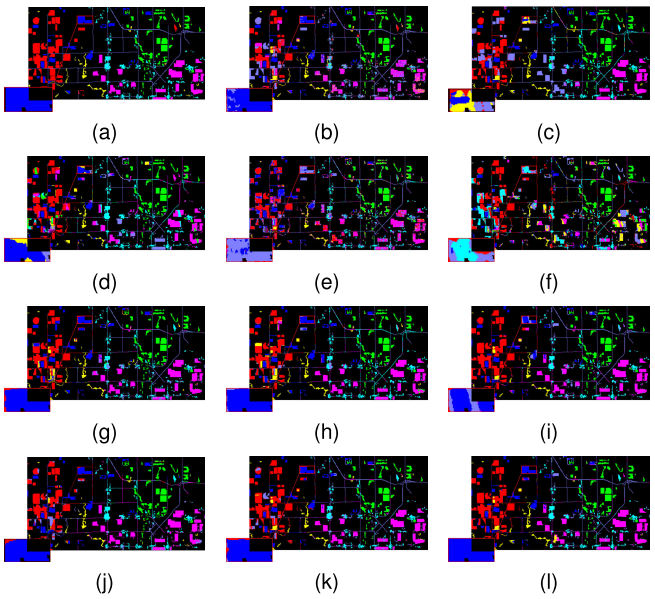



Fig. 17. Classification maps on Kansas under the 5-shot setting. (a) Ground truth, (b) SVM, (c) SSRN, (d) HSI-SNN, (e) DCFSL, (f) CMFSL, (g) Gia-CFSL, (h) HyMuT, (i) MRLF, (j) SGPL, (k) TEFSL, and (l) SpikeFSL.

TABLE XV
OA (%) COMPARISON UNDER DIFFERENT TRAINING ITERATIONS

| | | OA (1,000 Iterations→10,000 Iterations) | Improvement (ΔOA) |
|---|---|---|---|
| CMFSL | IP | 65.71%→68.26% | 2.55% |
| | HT | 77.02%→81.25% | 4.23% |
| | HH | 63.48%→73.36% | 9.88% |
| | Kansas | 53.74%→61.55% | 7.81% |
| Gia-CFSL | IP | 59.05%→64.62% | 5.57% |
| | HT | 75.64%→78.39% | 2.75% |
| | HH | 70.07%→76.63% | 6.56% |
| | Kansas | 73.53%→81.91% | 8.38% |
| HyMuT | IP | 56.57%→68.51% | 11.94% |
| | HT | 78.07%→80.66% | 2.59% |
| | HH | 72.28%→79.28% | 7.00% |
| | Kansas | 72.88%→78.76% | 5.88% |
| MRLF | IP | 66.90%→70.18% | 3.28% |
| | HT | 77.02%→79.79% | 2.77% |
| | HH | 74.16%→79.34% | 5.18% |
| | Kansas | 77.96%→79.30% | 1.34% |
| SGPL | IP | 59.32%→64.68% | 5.36% |
| | HT | 62.53%→74.93% | 12.40% |
| | HH | 80.03%→82.88% | 2.85% |
| | Kansas | 74.81%→78.66% | 3.85% |
| TEFSL | IP | 58.13%→66.84% | 8.71% |
| | HT | 79.67%→81.14% | 1.47% |
| | HH | 76.98%→80.54% | 3.56% |
| | Kansas | 75.43%→80.53% | 5.10% |
| SpikeFSL(Ours) | IP | **72.04%→72.62%** | **0.58%** |
| | HT | **82.86%→84.69%** | **1.83%** |
| | HH | **83.73%→83.83%** | **0.10%** |
| | Kansas | **82.31%→82.70%** | **0.39%** |

SpikeFSL can achieve effective target-domain generalization with substantially fewer training iterations.

### H. Computational Complexity Analysis

This section analyzes the computational complexity of SpikeFSL by comparing its FLOPs, running time required for effective target-domain generalization, parameter count, and OA with those of CMFSL, Gia-CFSL, HyMuT, MRLF, SGPL,

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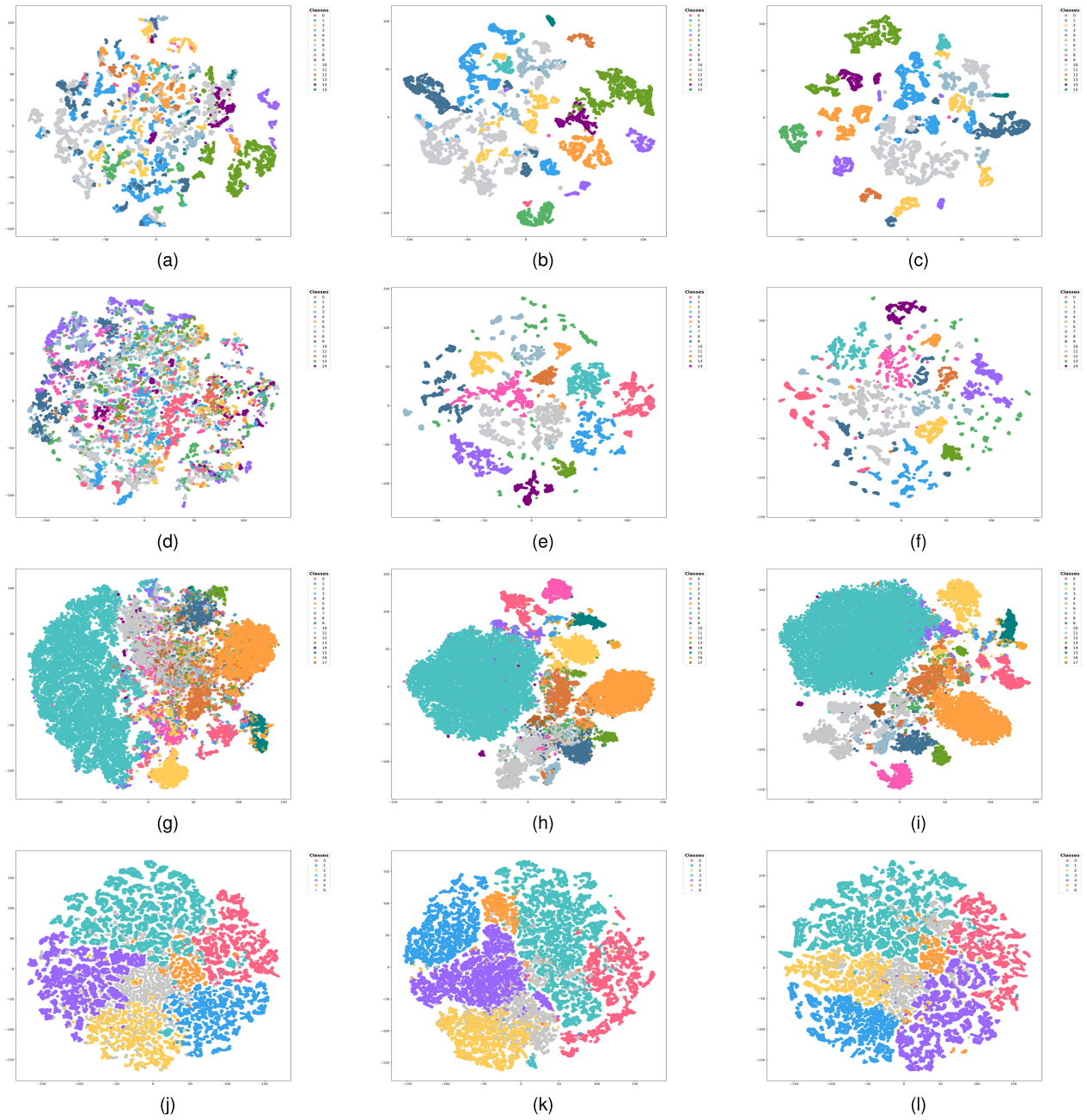



Fig. 18. t-SNE visualizations of feature distributions produced by SGPL, TEFSL, and the proposed SpikeFSL on four target-domain datasets. (a) SGPL on IP, (b) TEFSL on IP, (c) SpikeFSL on IP, (d) SGPL on HT, (e) TEFSL on HT, (f) SpikeFSL on HT, (g) SGPL on HH, (h) TEFSL on HH, (i) SpikeFSL on HH, (j) SGPL on kansas, (k) TEFSL on kansas, and (l) SpikeFSL on kansas.

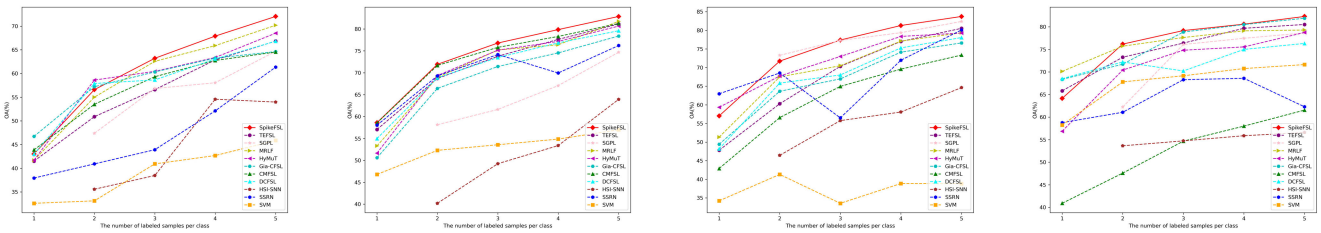



Fig. 19. Impact of the number of labeled samples per class on OA (%) across four target-domain datasets. (a) IP, (b) HT, (c) HH, and (d) Kansas.

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TABLE XVI
COMPARATIVE COMPUTATIONAL COMPLEXITY AND PERFORMANCE OF SPIKEFSL AND OTHER FSL METHODS

| | | IP | HT | HH | Kansas |
|---|---|---|---|---|---|
| CMFSL | FLOPs | 27.13 M | 27.13 M | 27.13 M | 27.13 M |
| | Time | 927.56 s | 801.31 s | 1072.46 s | 442.84 s |
| | Parameters | 613.991 K | 613.991 K | 613.991 K | 613.991 K |
| | OA | 68.26% | 81.25% | 73.36% | 61.55% |
| Gia-CFSL | FLOPs | 43.31 M | 42.86 M | 43.88 M | 44.19 M |
| | Time | 2187.33 s | 1763.66 s | 2004.58 s | 2404.62 s |
| | Parameters | 205.392 K | 199.792 K | 212.392 K | 216.192 K |
| | OA | 64.62% | 78.39% | 76.63% | 81.91% |
| HyMuT | FLOPs | 43.17 M | 42.71 M | 43.73 M | 44.04 M |
| | Time | 3984.33 s | 3478.04 s | 3840.12 s | 2184.33 s |
| | Parameters | **57.696 K** | **51.935 K** | **65.018 K** | **67.047 K** |
| | OA | 68.51% | 80.66% | 79.28% | 78.76% |
| MRLF | FLOPs | 49.10 M | 48.65 M | 49.67 M | 47.01 M |
| | Time | 1045.67 s | 873.97 s | 959.10 s | 856.79 s |
| | Parameters | 1532.624 K | 1527.024 K | 1539.624 K | 1543.424 K |
| | OA | 70.18% | 79.79% | 79.34% | 79.30% |
| SGPL | FLOPs | 149.98 M | 149.98 M | 149.98 M | 149.98 M |
| | Time | 1426.04 s | 1427.58 s | 1430.95 s | 1422.19 s |
| | Parameters | 373.056 K | 373.056 K | 373.056 K | 373.056 K |
| | OA | 64.68% | 74.93% | 82.88% | 78.66% |
| TEFSL | FLOPs | 58.87 M | 57.96 M | 60.00 M | 60.62 M |
| | Time | 1599.48 s | 1743.52 s | 1889.40 s | 766.43 s |
| | Parameters | 966.896 K | 961.296 K | 973.896 K | 977.696 K |
| | OA | 66.84% | 81.14% | 80.54% | 80.53% |
| SpikeFSL(Ours) | FLOPs | **1.99 M** | **1.53 M** | **2.56 M** | **2.87 M** |
| | Time | **293.94 s** | **272.13 s** | **358.45 s** | **226.01 s** |
| | Parameters | 171.572 K | 160.211 K | 185.894 K | 191.723 K |
| | OA | **72.04%** | **82.86%** | **83.73%** | **82.31%** |

and TEFSL, as summarized in Table XVI. Taking the IP dataset as an example, SpikeFSL requires only 1.99 M FLOPs and 293.94 s for effective target-domain generalization, which are significantly lower than those of all competing methods. HyMuT has the smallest parameter count of 57.696 K, but its FLOPs and running time required for effective target-domain generalization reach 43.17 M and 3984.33 s, respectively, which are much higher than those of SpikeFSL. This result suggests that the sparse, event-driven computation of SNNs contributes to the substantially lower FLOPs of SpikeFSL. Meanwhile, although SpikeFSL does not have the smallest parameter count among all compared methods, it remains compact and uses substantially fewer parameters than most other methods, such as MRLF with 1532.624 K parameters and TEFSL with 966.896 K parameters. Despite its relatively lightweight architecture, SpikeFSL achieves the highest OA, e.g., 72.04% on the IP dataset. Similar trends are also observed on the HT, HH, and Kansas datasets. Overall, these results indicate that SpikeFSL achieves a favorable balance among accuracy, computational complexity, and efficiency for target-domain generalization.

## V. CONCLUSION

This article proposes the SpikeFSL framework for cross-domain HSI classification, which leverages the sparse and event-driven characteristics of SNNs. Through the joint design of three tightly coupled components, SpikeFSL enables efficient and generalizable feature learning under data-scarcity and domain-shift conditions. Specifically, the SSTGSE module converts HSI pixels into SNN-compatible spiking sequences with spike-driven attentional selectivity, enabling effective joint modeling of spectral, spatial, and temporal information. Based on this encoding, the LSCFEN combines spiking convolutional blocks with 1-D spectral convolutions, achieving a balance between computational efficiency and discriminative feature extraction. Furthermore, the proposed IFPR strategy mitigates the feature-space inconsistency between SNN and CNN paradigms, facilitates the complementary fusion of their respective strengths, and improves cross-domain generalization performance via prototype refinement focused on hard-negative samples. Extensive experiments on multiple benchmark datasets verify that SpikeFSL outperforms state-of-the-art FSL methods in classification accuracy while significantly reducing computational cost, demonstrating its potential for deployment on resource-constrained remote sensing platforms. Nevertheless, the current framework still relies on a hybrid CNN–SNN architecture, which introduces additional model parameters and limits the exploration of fully spiking-based representations. Future work will focus on developing lightweight architectures based on purely SNN models, as well as exploring adaptive mechanisms for hyperparameter tuning, to better exploit their event-driven advantages and address more complex cross-domain HSI classification scenarios.

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